Asymmetric Information in Labor Contracts: Evidence from an Online Experiment

Daniel Herbst

University of Arizona

November 2024

Insuring Labor Product

- Most workers' labor product combines effort and risk
- Would likely value **insurance** against that risk
- What would be the premium for a contract that...
  - Pays restaurant servers when they're stiffed on the tip?
  - Pays lawyers when they lose a case?
  - Pays academics when they're rejected from journals?

- Employers provide *implicit insurance* through <u>**fixed wages**</u>

Piece-rate Pay

- Paid by unit of output
- Worker sells labor product at market price
- Examples:
  - Rideshare driver paid per mile
  - Self-employed freelance photographer

Fixed Wages

- Paid by unit of time
- Worker sells *claim* on labor product
- Examples:
  - Limo driver paid per hour
  - Staff photographer with annual salary

← Earnings Risk

Insurance →

Hourly wage + tips

Base pay + commission

Annual salary + bonus

Are these contracts socially optimal?

Piece-rate Pay

- Paid by unit of output
- Worker sells labor product at market price
- Examples:
  - Rideshare driver paid per mile
  - Self-employed freelance photographer

Fixed Wages

- Paid by unit of time
- Worker sells *claim* on labor product
- Examples:
  - Limo driver paid per hour
  - Staff photographer with annual salary

Earnings Risk Insurance

Asymmetric information ⇒ market distortions:

- **Moral hazard:** fixed wage induces less effort
- **Adverse selection:** less productive workers sort into fixed wages

⇒ Suboptimal wage contracts

Public policies can mitigate these distortions:

- Hourly wage subsidies
- Employment classification rules
- Portable benefits programs
- Minimum wage

⇒ Promote implicit insurance in wage contracts

How do moral hazard and adverse selection influence wage contracts?

This Paper:

1. Experimentally identify treatment effects and selection into hourly pay
   - Two-stage RCT separates <u>moral hazard</u>, <u>adverse selection</u>, and <u>wage effects</u>
     1. Offer workers a *choice* between randomized hourly wage or piece rate
     2. Randomly increase hourly workers' wages to match highest wage offer
   - Find evidence of both moral hazard and adverse selection into hourly wages
     - MH ≡ <u>Treatment effect</u>: take-up of hourly contract ⇒ output ↓ by 6.3% relative to the mean
     - AS ≡ <u>Selection into treatment</u>: wage offer ↑ by 10% ⇒ marginal worker's productivity ↑ by 1.4%

This Paper:

1. Experimentally identify treatment effects and selection into hourly pay

2. Develop model of fixed-wage labor markets under asymmetric info
    - Clarifies roles of AS and MH in shaping equilibrium, allowing for monitoring costs
    - Equilibrium and efficient shares of hourly work are determined by three curves:
        1. $\bar{w}(\theta)$: **reservation wage** for an hourly position at quantile θ
        2. $MV(\theta)$: **marginal value** of output among workers at a given reservation wage
        3. $AV(\theta)$: **average value** of output among workers with lower reservation wage

This Paper:

1. Experimentally identify treatment effects and selection into hourly pay

2. Develop model of fixed-wage labor markets under asymmetric info

3. Use MTE framework to estimate equilibrium and efficient hourly wages
   - Let experimental wage offers serve as **instrument** for hourly contract take-up ⇒
     - Hourly labor supply at given wage, $S(w)$ = propensity score
     - Marginal value under hourly pay, $MV_1(\theta)$ = potential outcome in treated state
     - Marginal value under piece rate, $MV_0(\theta)$ = potential outcome in untreated state

   ⇒ MTE of hourly contracts map directly into model

This Paper:

1. Experimentally identify treatment effects and selection into hourly pay

2. Develop model of fixed-wage labor markets under asymmetric info

3. Use MTE framework to estimate equilibrium and efficient hourly wages

4. Calculate DWL of inefficient wages and welfare impact of subsidies
    - Welfare loss of \$0.03 to \$0.05 per hour of labor
    - Wage subsidies $\leq$ \$1.00 $\Rightarrow$ Marginal Value of Public Funds (MVPF) $\in (0.95, 1.15)$

Existing Literature

- Asymmetric Information in Labor Markets
  - Spence (1973), Weiss (1980), Gibbons and Katz (1991), Farber and Gibbons (1996), Shimer (2005), Pallais (2016), Marinescu and Wolthoff (2020)
- Selection and Incentive Effects of Wage Contracts
  - Lazear (2000), Shearer (2006), DellaVigna and Pope (2018), Kantarevic and Kralj (2016), Angrist et al. (2017), Brown and Andrabi (2021), Emanuel and Harrington (2024)
- Measuring Adverse Selection & Moral Hazard
  - Chiappori and Salanie (2000), Karlan and Zinman (2009), Einav, Finkelstein, Cullen (2010), Kowalski (2023), Herbst and Hendren (2024)
- Selection and Marginal Treatment Effects
  - Black et al. (2022), Mogstad et al. (2018), Huber (2013), Carneiro et al. (2011), Heckman and Vyctlacil (2005, 2007), Björklund and Moffitt (1987)

Outline

1. Experimental Design
2. Main Results
3. Model of Asymmetric Information
4. Estimates of Marginal Value and Welfare Loss
5. Optimal Wage Subsidies

Detecting Moral Hazard and Adverse Selection

Existing methods of testing for moral hazard and adverse selection:

1. Test for correlation between take-up and realized risk (Chiappori and Salanie 2000)
    - Cannot separate adverse selection from moral hazard
2. Compare take-up and risk across exogenous price changes in existing markets (Einav et al. 2010, Einav et al. 2013; Hackmann et al. 2012)
    - "Under-the-lamppost" problem: Cannot observe *unraveled* contracts, which are unprofitable
3. Construct *hypothetical* contract choices from surveys on subjective beliefs (Herbst and Hendren 2024; Hendren 2013, 2017)
    - Strong parametric assumptions to predict choices across "missing" contracts

My strategy: if I can't observe missing contracts, offer them myself!

- Advantage #1: I can experimentally randomize contract prices (i.e., wages)
- Advantage #2: I can offer unraveled (i.e., unprofitable) contracts

Detecting Moral Hazard and Adverse Selection

- Suppose worker i is offered a choice:
  - Remain on piece rate $(H_i = 0)$
  - Switch to an hourly wage $(H_i = 1)$
- Potential outcomes:

  Y_{1i}: Worker i's potential output under hourly wage

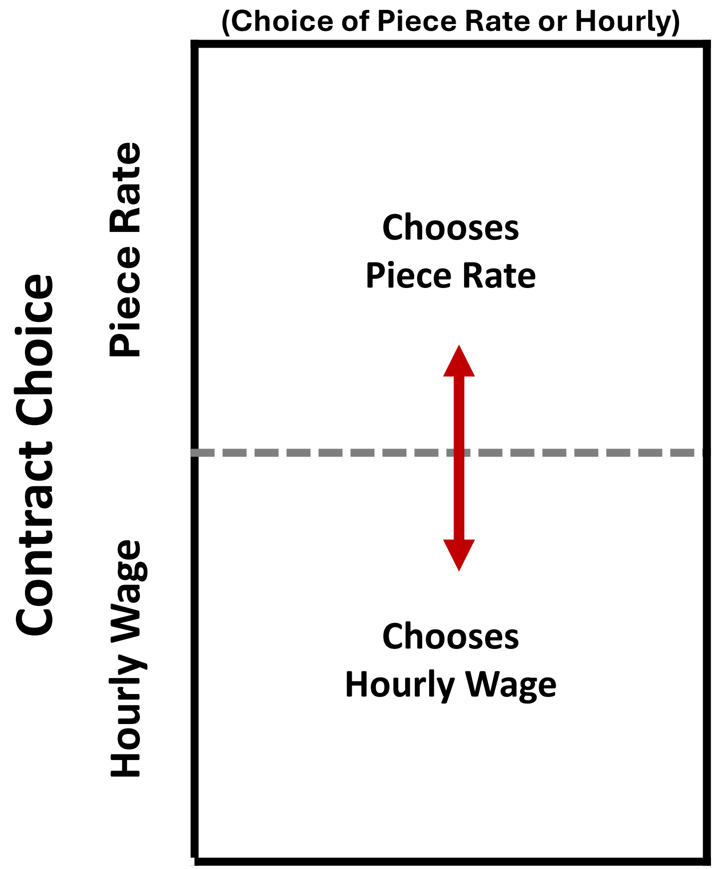
  Y_{0i}: Worker i's potential output under piece rate

$$Y_i = H_i Y_{1i} + (1 - H_i) Y_{0i}$$

- Comparing across self-selected groups combines treatment and selection:

$$E[Y_i | H_i = 1] - E[Y_i | H_i = 0] =$$

$$\underbrace{E[Y_{1i} - Y_{0i} | H_i = 1]}_{\text{Treatment on the Treated}} + \underbrace{E[Y_{0i} | H_i = 1] - E[Y_{0i} | H_i = 0]}_{\text{Selection on } Y_0}$$

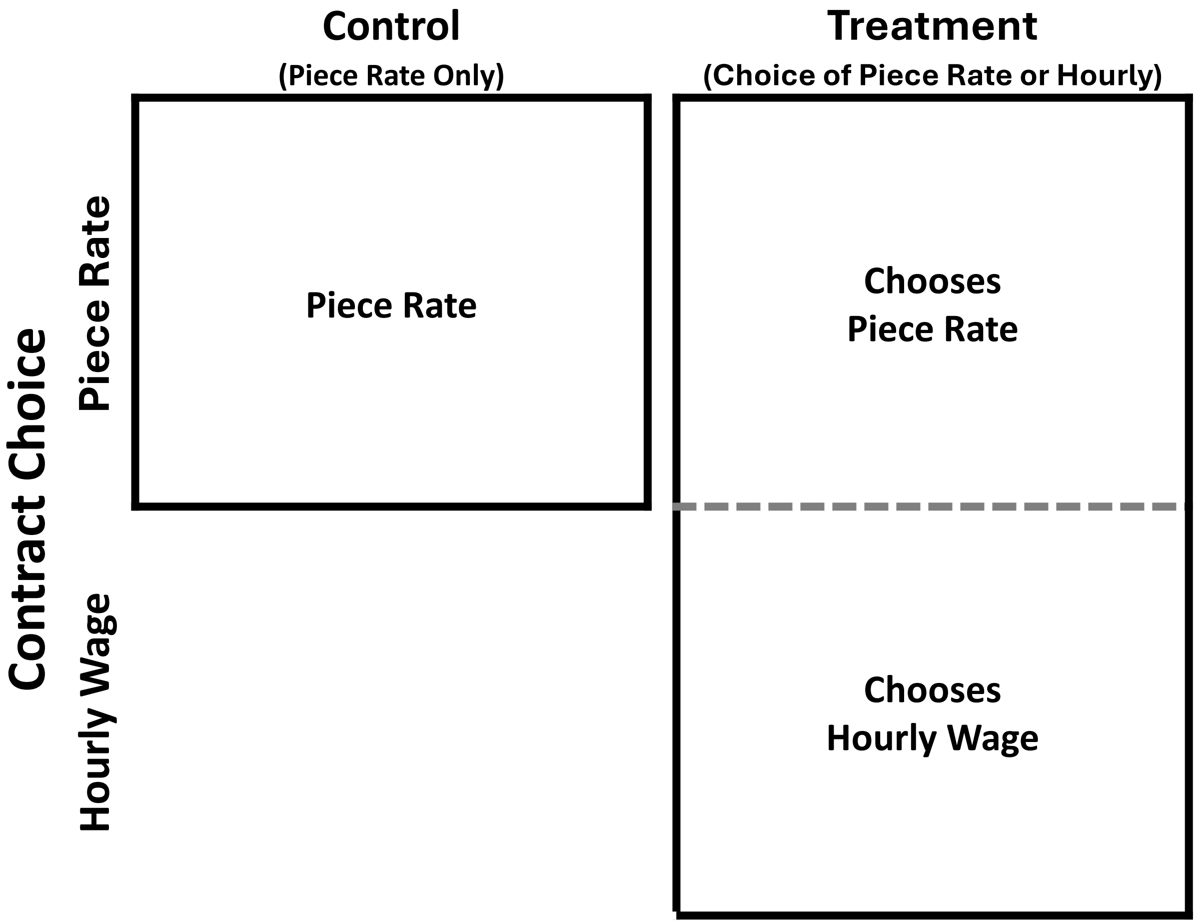


Experimental Design: Single Wage Offer

<u>Simplified Experimental Design</u>

Randomize workers into two offer sets:

- **Treated workers ($W_i = 1$):** offered choice
  - Remain on piece rate ($H_i = 0$)
  - Switch to an hourly wage ($H_i = 1$)
- **Control workers ($W_i = 0$):** no hourly offer
  - Remain on piece rate ($H_i = 0$)



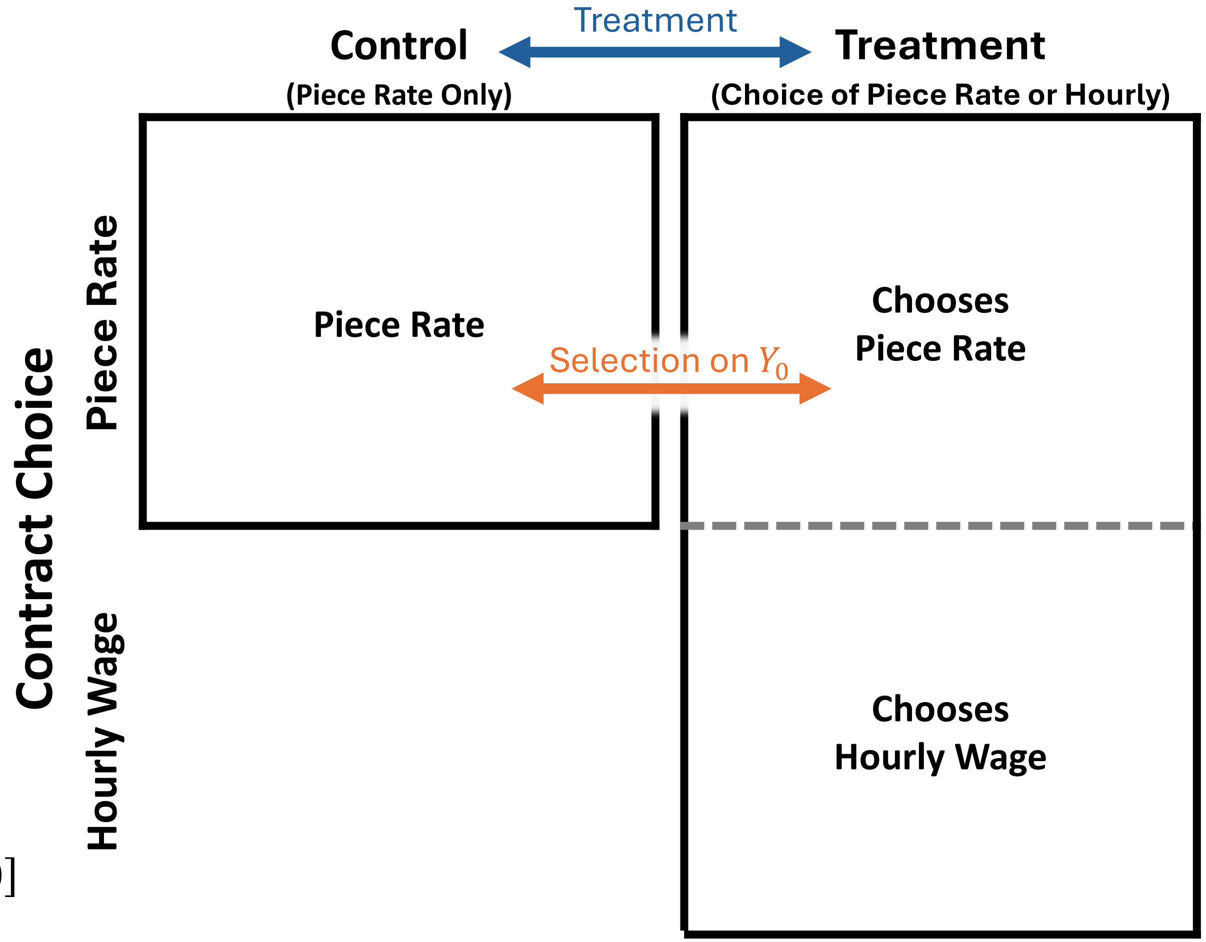
Experimental Design: Single Wage Offer

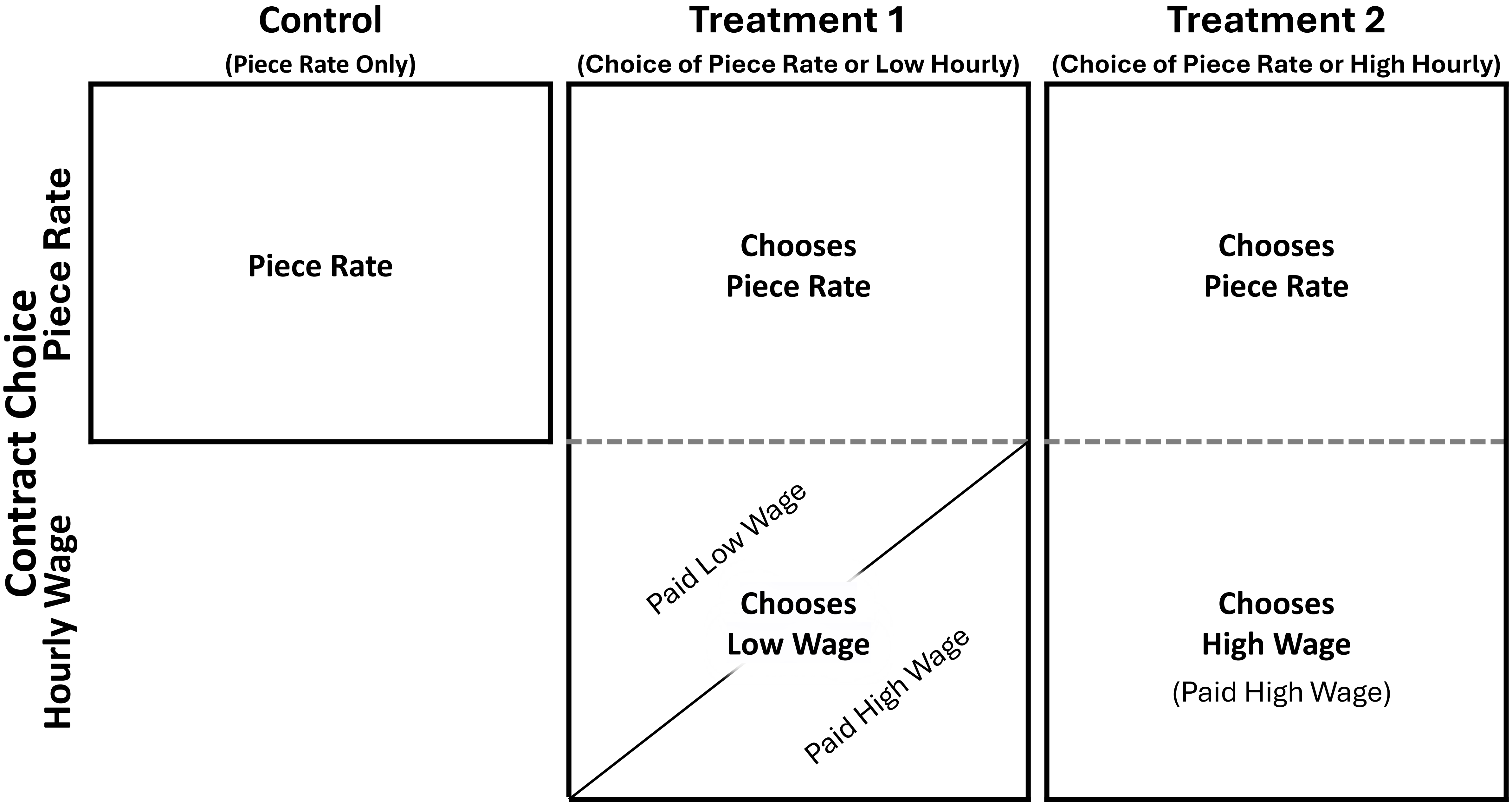
Comparing between workers facing different hourly offers identifies ***treatment effect*** of hourly wages:

$$\frac{E[Y_i|W_i=1]-E[Y_i|W_i=0]}{\Pr(H_i=1|W_i=1)} = \overset{\text{Moral Hazard}}{\underbrace{E[Y_{1i}-Y_{0i}|H_i=1]}_{\substack{\text{Local Average Treatment Effect} \\ \text{(LATE)}}}}$$

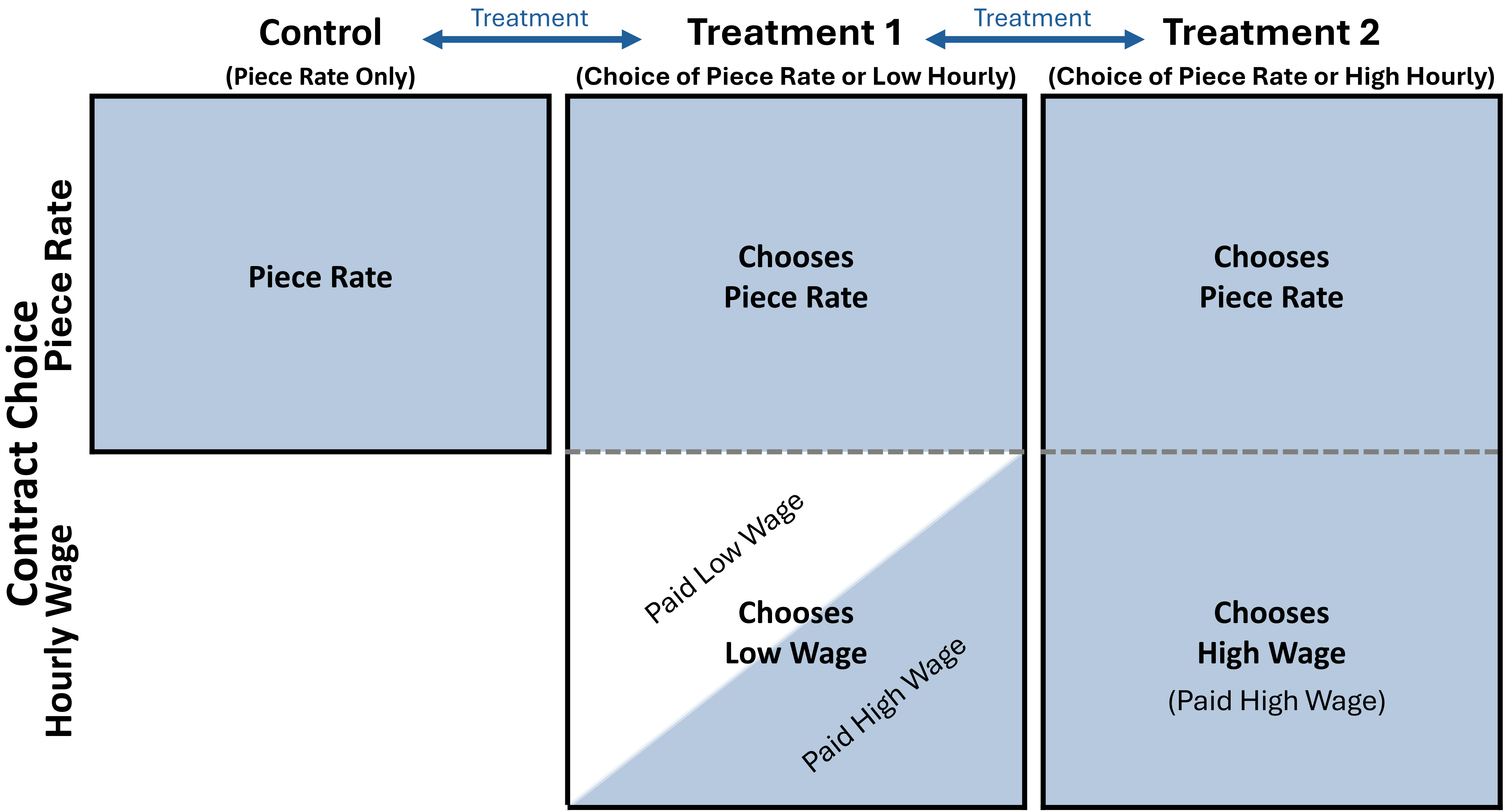
Comparing piece-rate workers who faced different offer sets identifies ***selection into treatment:***

$$\frac{E[Y_i|H_i=0,W_i=1]-E[Y_i|W_i=0]}{\Pr(H_i=1|W_i=1)} = \overset{\text{Adverse Selection}}{\underbrace{E[Y_{0i}|H_i=1]-E[Y_{0i}|H_i=0]}_{\text{Selection on } Y_0}}$$

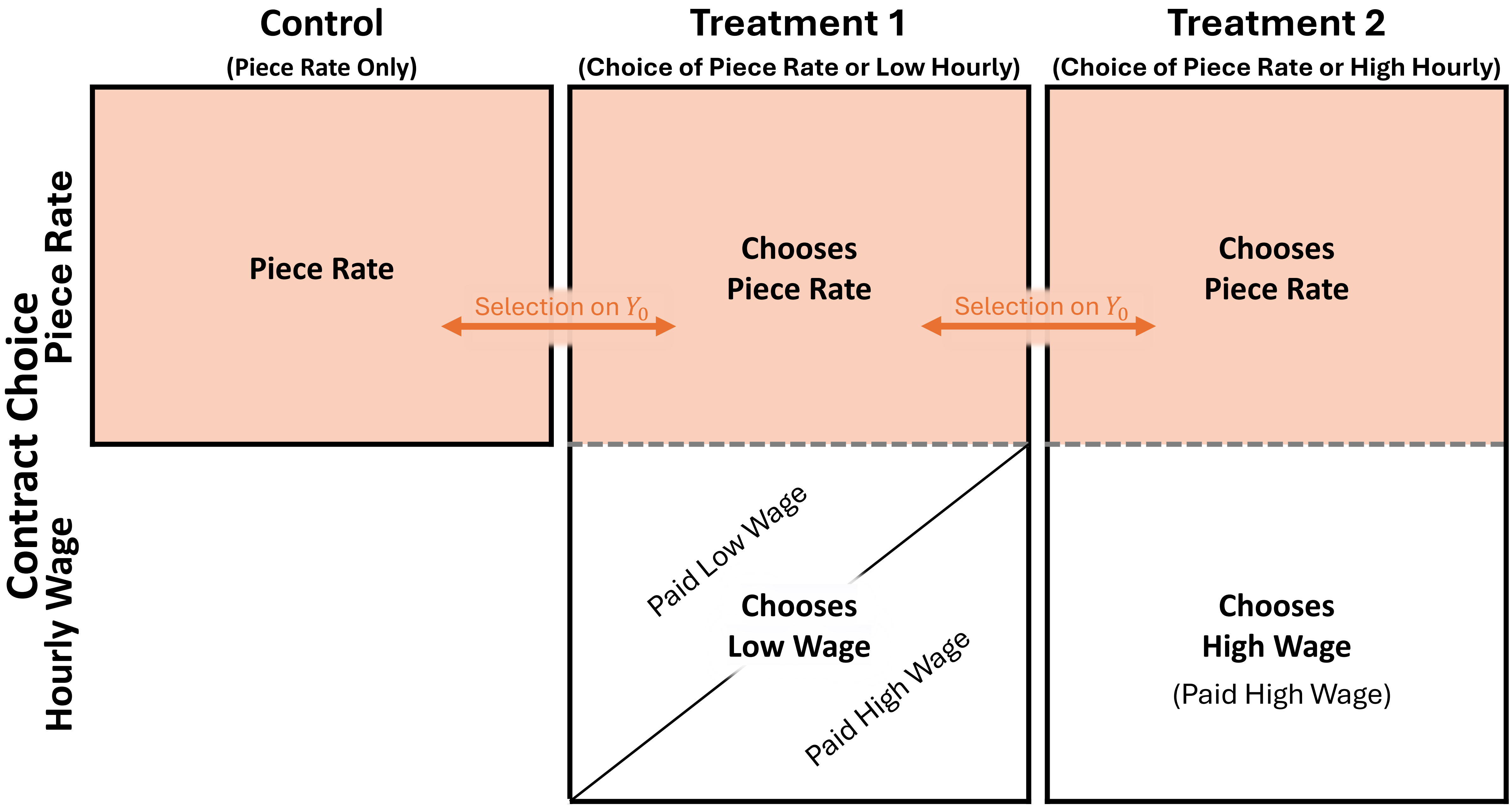




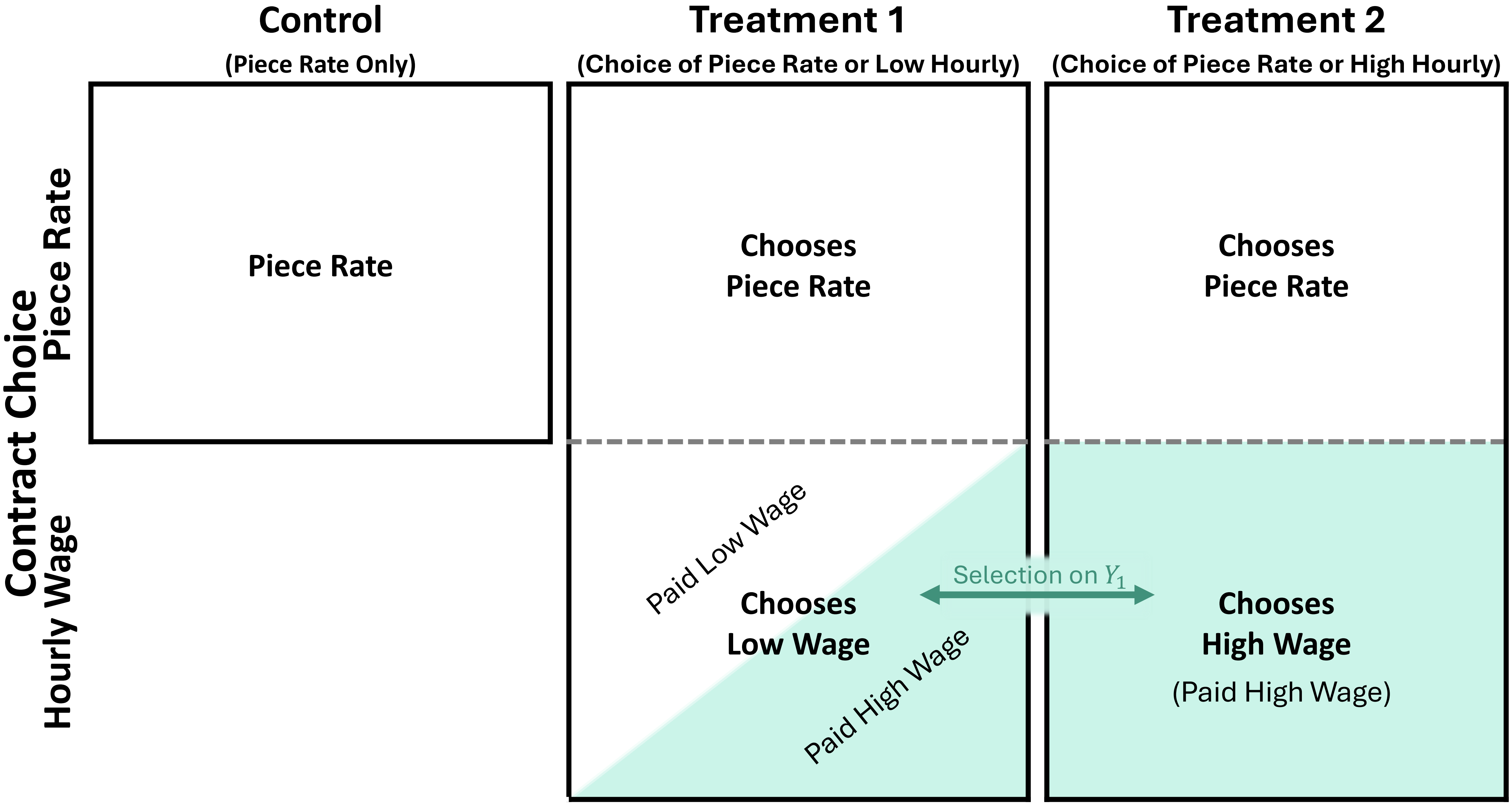
Formal Expression



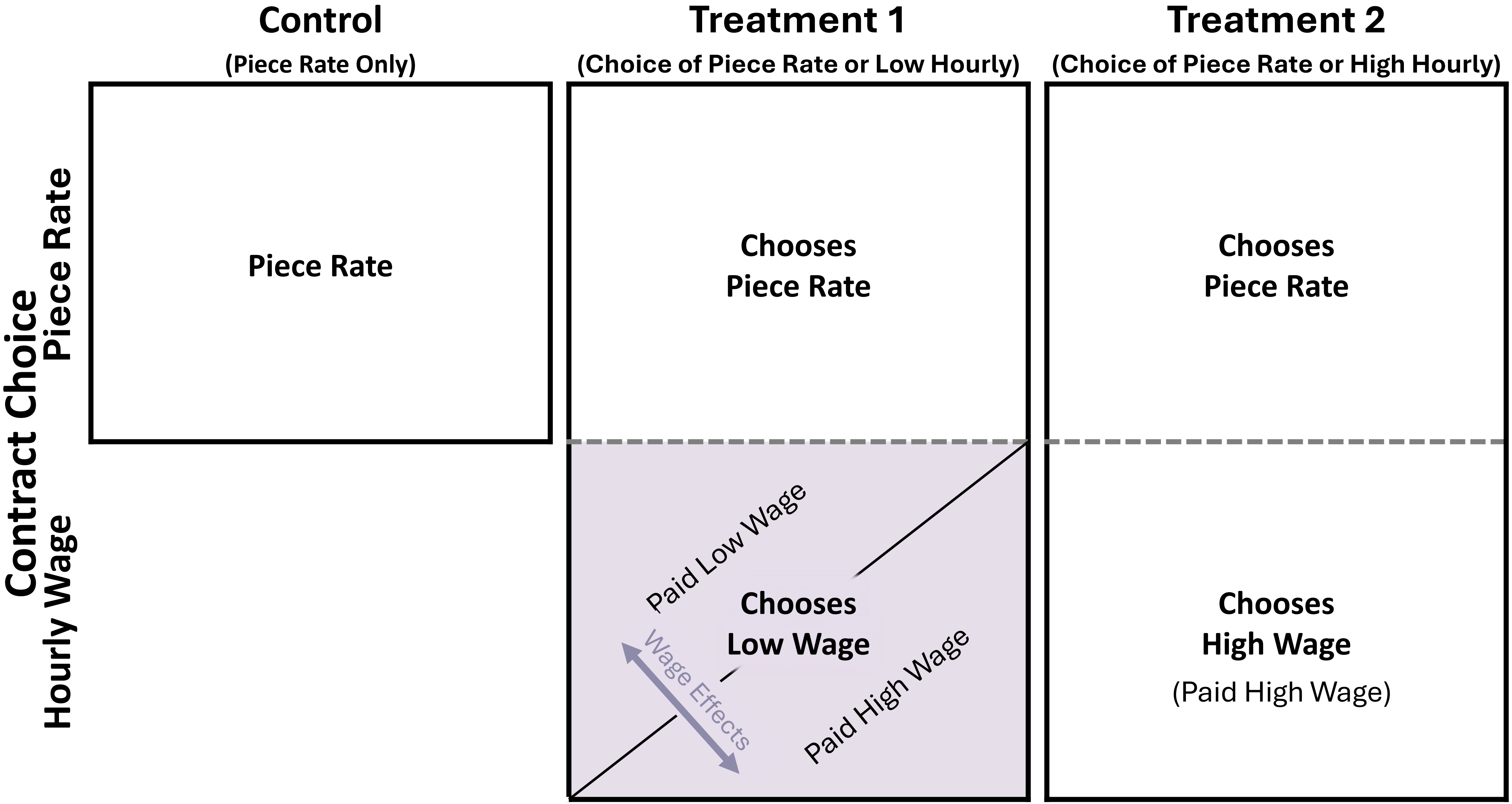
Formal Expression



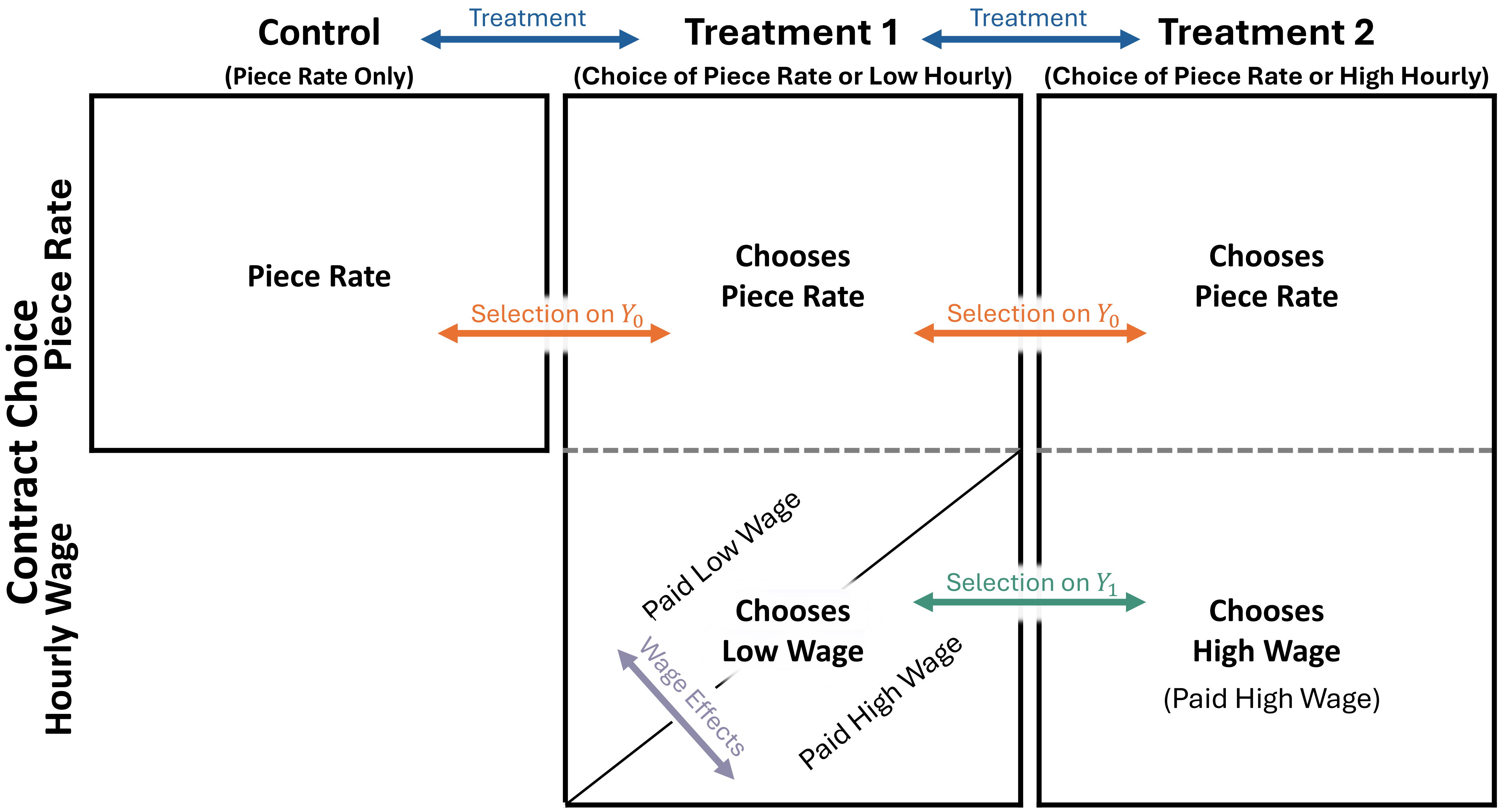
Formal Expression



Formal Expression

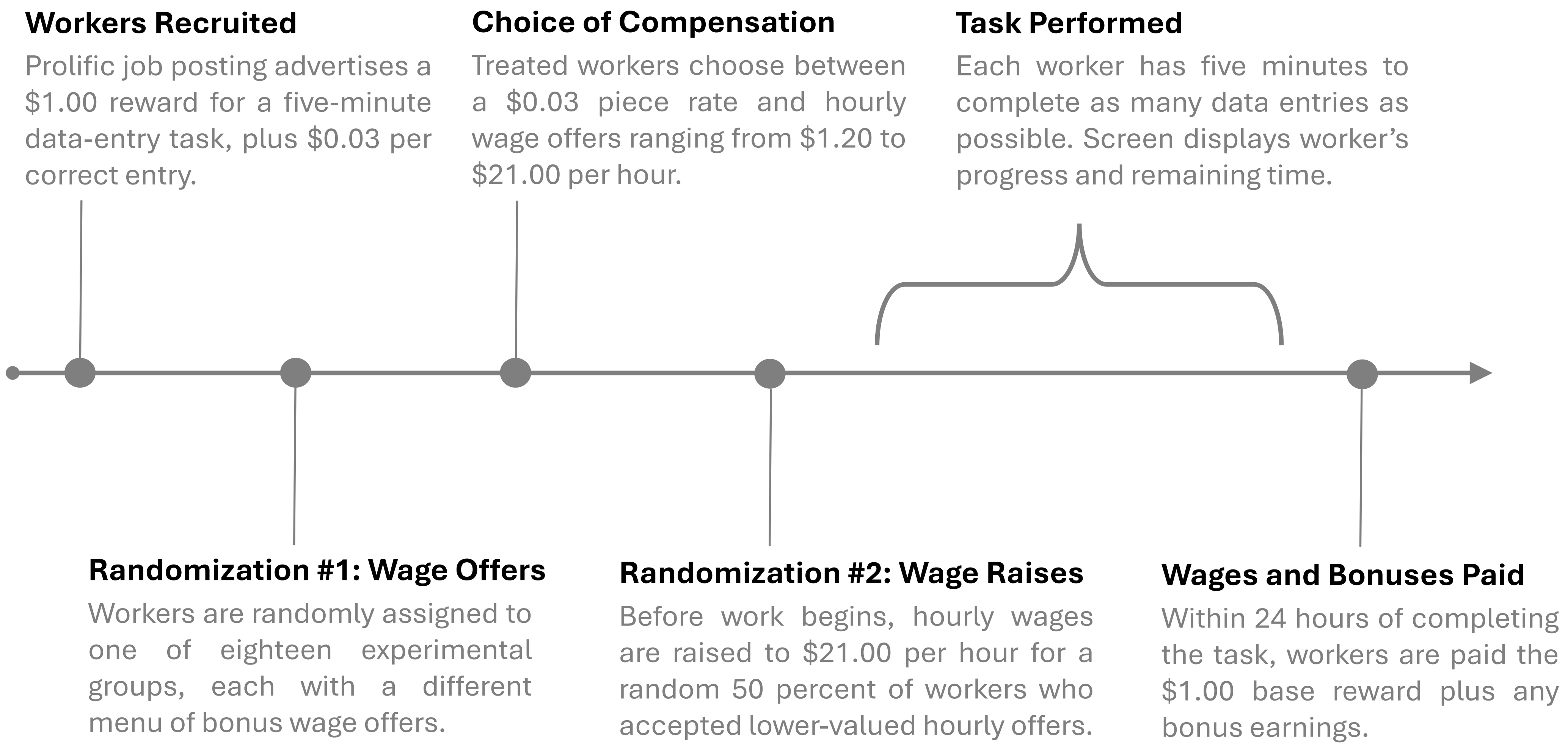


Formal Expression



Formal Expression

Experimental Procedure



Experimental Procedure

Workers Recruited

Prolific job posting advertises a $1.00 reward for a five-minute data-entry task, plus $0.03 per correct entry.

- Posting screens for
  - Approval rating ≥98%
  - Number of approved tasks ≥10
  - Located within US
- In practice, most users who see the posting accept the job.
- Experiment takes place over ten waves of ≈300 workers

$1.00 • $12.00/hr | 5 mins | 300 places | Writing

You will be shown a series of handwritten sentences for 5 minutes. Your task is to type each sentence into the corresponding text box.

*You can earn an additional $0.03 in bonus compensation for each **correctly** typed sentence.* Any bonus payments will be deposited within 24 hours of completion. You must reach the end of the 5-minute task to receive credit for your submission.

Experimental Procedure

Workers Recruited

Prolific job posting advertises a $1.00 reward for a five-minute data-entry task, plus $0.03 per correct entry.

Randomization #1: Wage Offers

Workers are randomly assigned to one of eighteen experimental groups, each with a different menu of bonus wage offers.

Hourly Wage Offer	Piece-Rate Offer	Number of Participants
No Hourly Offer	$0.03 per sentence	302
$1.20/hr	$0.03 per sentence	300
$1.80/hr	$0.03 per sentence	101
$2.40/hr	$0.03 per sentence	103
$3.00/hr	$0.03 per sentence	304
$3.60/hr	$0.03 per sentence	100
$4.20/hr	$0.03 per sentence	99
$4.80/hr	$0.03 per sentence	101
$5.40/hr	$0.03 per sentence	101
$6.00/hr	$0.03 per sentence	305
$7.20/hr	$0.03 per sentence	100
$8.40/hr	$0.03 per sentence	102
$9.60/hr	$0.03 per sentence	101
$10.80/hr	$0.03 per sentence	100
$12.00/hr	$0.03 per sentence	305
$15.00/hr	$0.03 per sentence	100
$18.00/hr	$0.03 per sentence	102
$21.00/hr	$0.03 per sentence	304
		Total: 3030

Experimental Procedure

Workers Recruited

Prolific job posting advertises a $1.00 reward for a five-minute data-entry task, plus $0.03 per correct entry.

Randomization #1: Wage Offers

Workers are randomly assigned to one of eighteen experimental groups, each with a different menu of bonus wage offers.

Choice of Compensation

Treated workers choose between a $0.03 piece rate and hourly wage offers ranging from $1.20 to $21.00 per hour.

Before you begin the task, we'd like to offer you a choice of how to receive your bonus payment. Please select your preferred method of compensation from the options below:

Get paid a flat bonus of $1.00. ○	Get paid $0.03 for each sentence you correctly complete. ○

Experimental Procedure

Workers Recruited

Prolific job posting advertises a $1.00 reward for a five-minute data-entry task, plus $0.03 per correct entry.

Randomization #1: Wage Offers

Workers are randomly assigned to one of eighteen experimental groups, each with a different menu of bonus wage offers.

Choice of Compensation

Treated workers choose between a $0.03 piece rate and hourly wage offers ranging from $1.20 to $21.00 per hour.

Randomization #2: Wage Raises

Before work begins, hourly wages are raised to $21.00 per hour for a random 50 percent of workers who accepted lower-valued hourly offers.

For performing this task, you will receive **$1.75**, plus your chosen **flat bonus of $1.00**.

Your **total compensation will be $2.75**.

Experimental Procedure

Workers Recruited

Prolific job posting advertises a $1.00 reward for a five-minute data-entry task, plus $0.03 per correct entry.

Randomization #1: Wage Offers

Workers are randomly assigned to one of eighteen experimental groups, each with a different menu of bonus wage offers.

Choice of Compensation

Treated workers choose between a $0.03 piece rate and hourly wage offers ranging from $1.20 to $21.00 per hour.

Randomization #2: Wage Raises

Before work begins, hourly wages are raised to $21.00 per hour for a random 50 percent of workers who accepted lower-valued hourly offers.

Task Performed

Each worker has five minutes to complete as many data entries as possible. Screen displays worker's progress and remaining time.

Workers face threat of rejection ⇒

- Don't receive any earnings
- Reputational damage (approval rate ↓)

You will be shown a series of handwritten sentences. On each page, your task is to type the sentence into the text box below.

Here is one example of a completed sentence:

The quick brown fox jumps over the lazy dog.

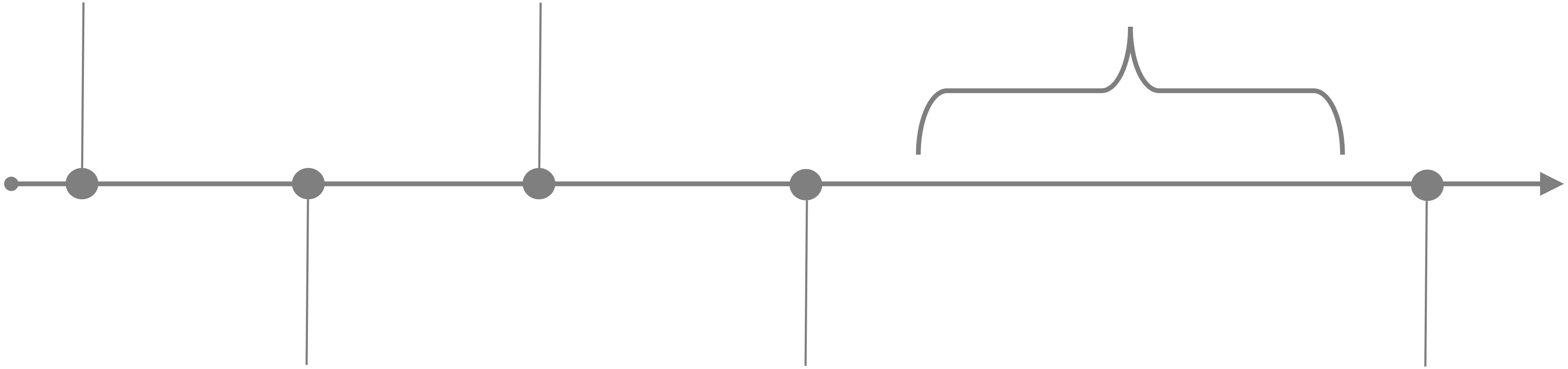
The quick brown fox jumps over the lazy dog.

Your answers should be as accurate as possible. Please be mindful of capitalization, spacing, and punctuation. When you've completed a sentence, click the "→" button to move on.

You will have <u>5 minutes</u> to complete as many sentences as you can. You cannot start over, and you can only perform this task once.

→

Experimental Procedure



Workers Recruited

Prolific job posting advertises a $1.00 reward for a five-minute data-entry task, plus $0.03 per correct entry.

Randomization #1: Wage Offers

Workers are randomly assigned to one of eighteen experimental groups, each with a different menu of bonus wage offers.

Choice of Compensation

Treated workers choose between a $0.03 piece rate and hourly wage offers ranging from $1.20 to $21.00 per hour.

Randomization #2: Wage Raises

Before work begins, hourly wages are raised to $21.00 per hour for a random 50 percent of workers who accepted lower-valued hourly offers.

Task Performed

Each worker has five minutes to complete as many data entries as possible. Screen displays worker's progress and remaining time.

Wages and Bonuses Paid

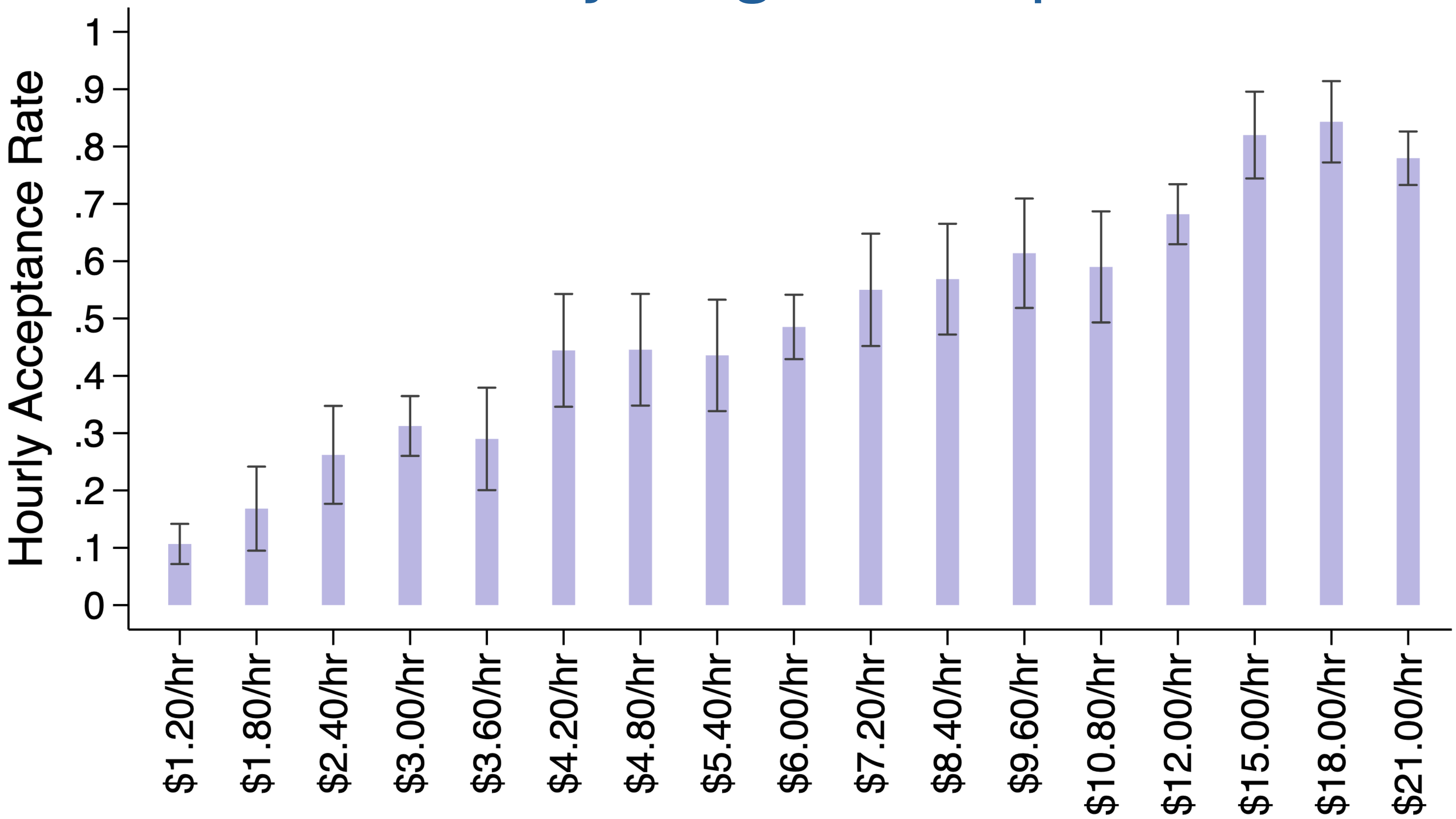
Within 24 hours of completing the task, workers are paid the $1.00 base reward plus any bonus earnings.

Outline

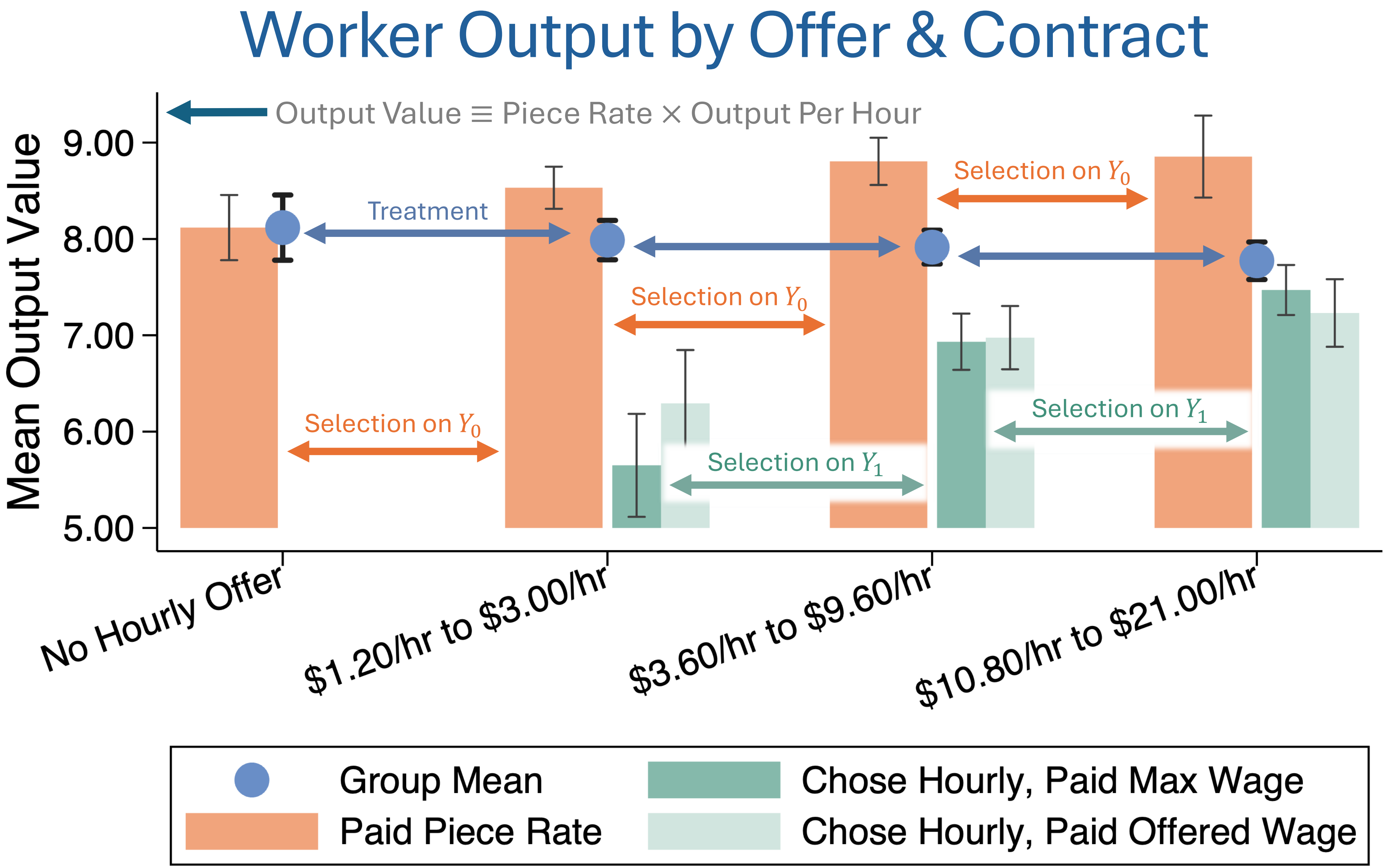
Summary Statistics

Category	Variable	Mean	SD
Panel A: Task Performance	Accepted Hourly Offer	0.438	0.496
	Completed Sentences	21.98	8.148
	Correct Sentences	17.79	9.360
	Output Value	7.912	2.933
	Finished	0.986	0.118
Panel B: Demographics & Employment	Age	37.23	12.18
	Female	0.643	0.479
	Minority	0.357	0.479
	Employed	0.685	0.465
	Student	0.187	0.390
	Number of Previous Tasks	1281.6	1746.4

Hourly Wage Take-up



Logit Estimates



Worker Output by Offer & Contract
Output Value ≡ Piece Rate × Output Per Hour
Mean Output Value
9.00
8.00
7.00
6.00
5.00
Treatment
Selection on Y_0
Selection on Y_0
Selection on Y_0
Selection on Y_1
Selection on Y_1
No Hourly Offer
$1.20/hr to $3.00/hr
$3.60/hr to $9.60/hr
$10.80/hr to $21.00/hr
Group Mean
Paid Piece Rate
Chose Hourly, Paid Max Wage
Chose Hourly, Paid Offered Wage

2SLS Estimates of Treatment Effects

	(1) Output Value	(2) Output Value	(3) Output Value	(4) Output Value
Accepted Hourly Offer	-0.506^{**}	-0.500^{**}	-0.488^{**}	-0.365^{**}
	(0.206)	(0.200)	(0.200)	(0.185)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
R-squared	0.037	0.080	0.096	0.232
N	3030	3030	3030	3030

$$\tilde{Y}_i = \delta H_i + \boldsymbol{\eta} \boldsymbol{X_i} + \epsilon_i$$

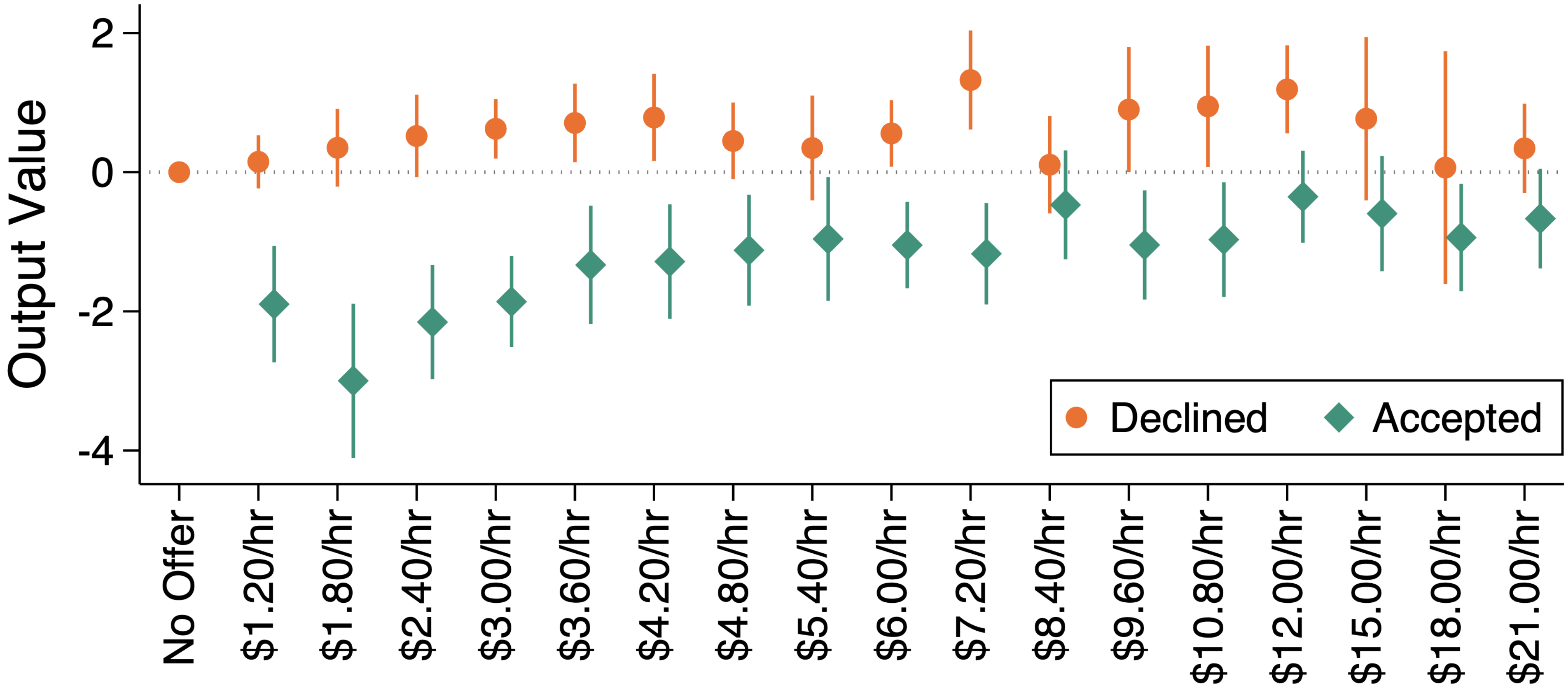
$$\tilde{Y}_i \equiv Y_i - \gamma H_i \times W_i^P$$

Selection by Log Wage Offer

	(1) Output Value	(2) Output Value	(3) Output Value	(4) Output Value
Accepted Hourly Offer	-2.598^{***}	-2.481^{***}	-2.439^{***}	-2.256^{***}
	(0.329)	(0.319)	(0.320)	(0.300)
Declined × Log Hourly Wage Offer	0.167^{*}	0.193^{**}	0.210^{**}	0.230^{***}
	(0.0960)	(0.0932)	(0.0925)	(0.0855)
Accepted × Log Hourly Wage Offer	0.621^{***}	0.570^{***}	0.568^{***}	0.501^{***}
	(0.116)	(0.112)	(0.113)	(0.104)
Accepted × Log Effective Hourly Wage	−0.0608	−0.0443	−0.0444	−0.0000829
	(0.122)	(0.118)	(0.118)	(0.110)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
R-squared	0.082	0.123	0.139	0.273
N	3030	3030	3030	3030

$$Y_i = \alpha H_i + \beta_0(1 - H_i)\times W_i + \beta_1 H_i \times W_i + \gamma H_i \times W_i^P + \boldsymbol{\xi} \boldsymbol{X_i} + \epsilon_i$$

Selection by Log Wage Offer



$$Y_i = \sum_{w \in \mathbb{W}} [(\beta_{w0}(1 - H_i) + \beta_{w1} H_i) \times \mathbf{1}\{W_i = w\}] + \gamma H_i \times W_i^P + \boldsymbol{\xi}\, \boldsymbol{X_i} + \epsilon_i$$

Summary

What have I shown so far?

- Consider two hourly wage offers, $W' > W$
- Let $\bar{w}_i \equiv$ **hourly reservation wage** (the lowest hourly wage worker i would accept)

Evidence of <u>Moral Hazard</u>

$$\underbrace{E[Y_{1i} - Y_{0i} | W < \bar{w}_i \leq W'] < 0}_{\text{Local Average Treatment Effect}}$$

Evidence of <u>Adverse Selection</u>

$$\underbrace{E[Y_{0i} | \bar{w}_i > W] < E[Y_{0i} | \bar{w}_i > W']}_{\text{Local Average Selection on } Y_0}$$

$$\underbrace{E[Y_{1i} | \bar{w}_i \leq W] < E[Y_{1i} | \bar{w}_i \leq W']}_{\text{Local Average Selection on } Y_1}$$

What are the implications for…

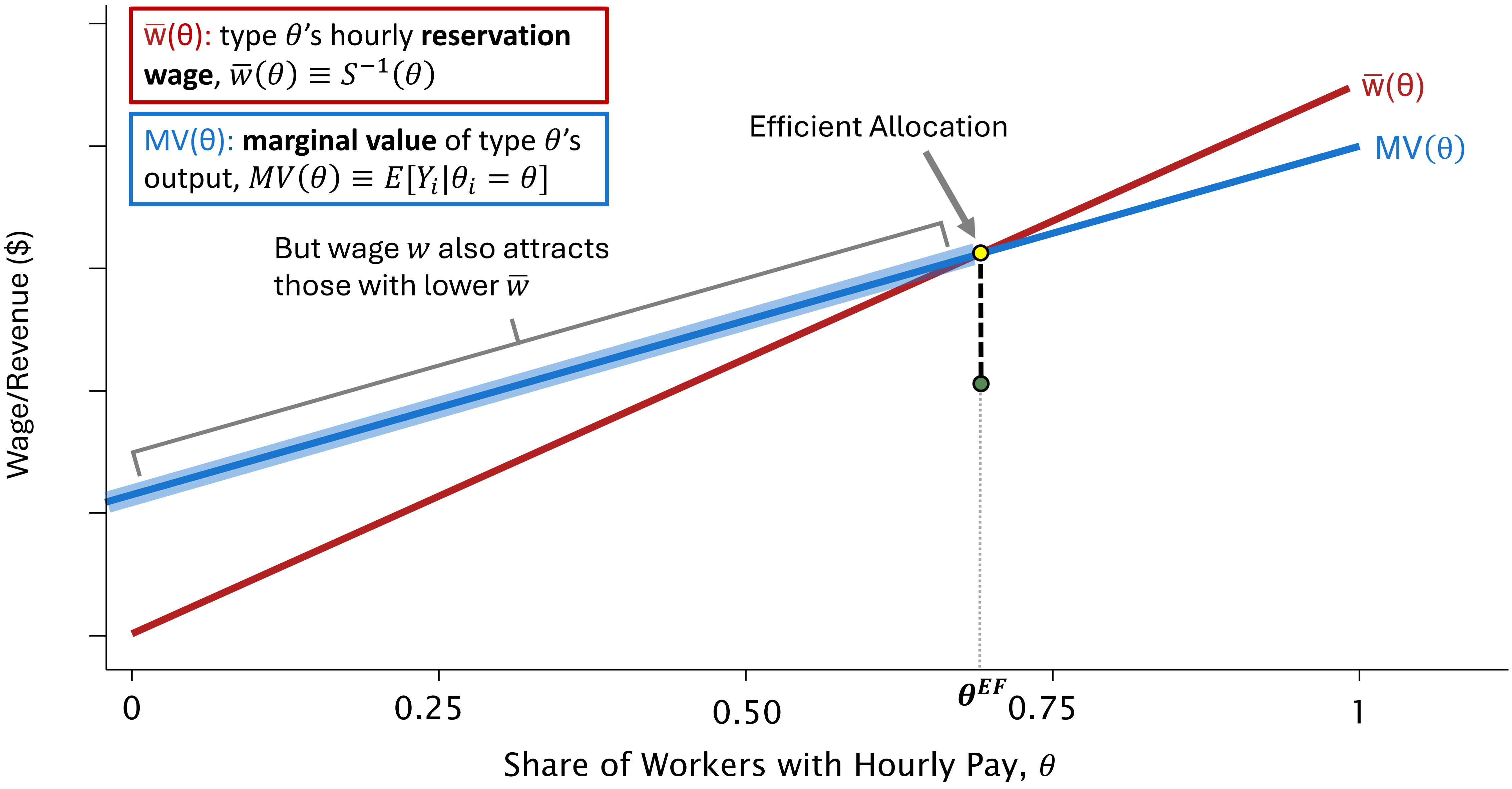
- Market equilibrium?
- Social welfare?
- Policy?

Outline

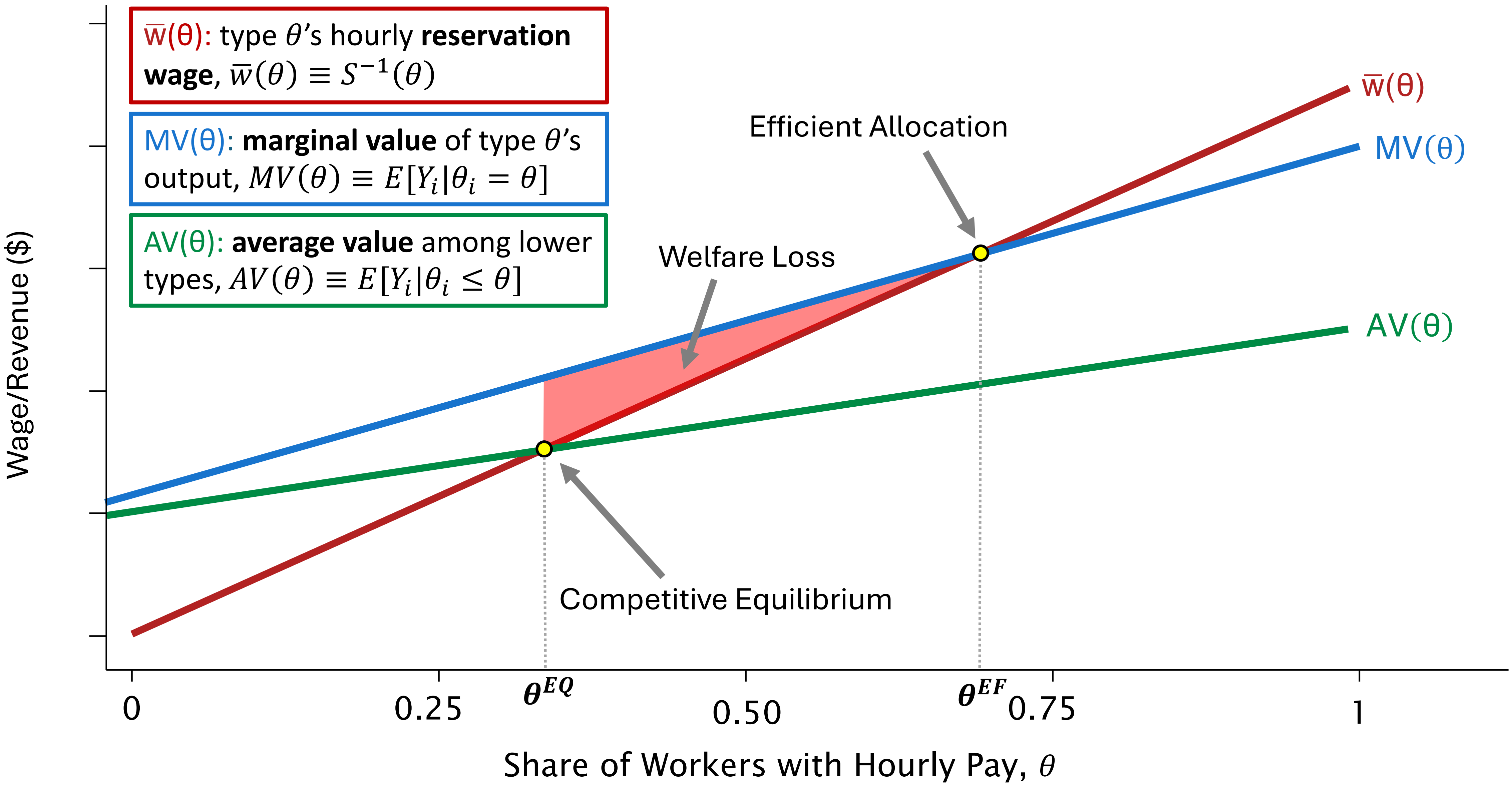
1. Experimental Design
2. Main Results
3. **Model of Asymmetric Information**
4. Estimates of Marginal Value and Welfare Loss
5. Optimal Wage Subsidies

Model

- Population of observationally equivalent (pre-screened) workers
- Each worker produces hourly output $q_i = f(\zeta_i, e_i, \nu_i)$
  - ζ_i: (unobserved) worker characteristics
  - e_i: individual effort
  - ν_i: random noise
- Value of worker i's output is $Y_i \equiv pq_i$, where $p \equiv$ price per unit q
- Firms have two options:
  1. Buy output at per-unit market price p (e.g., piece rate, freelance hire)
  2. Offer worker a flat hourly wage, w, for whatever they produce



$\overline{w}(\theta)$: type θ's hourly **reservation wage**, $\overline{w}(\theta) \equiv S^{-1}(\theta)$
MV(θ): **marginal value** of type θ's output, $MV(\theta) \equiv E[Y_i|\theta_i = \theta]$
Efficient Allocation
But wage w also attracts those with lower $\overline{w}$
$\overline{w}(\theta)$
MV(θ)
Wage/Revenue ($)
0
0.25
0.50
θ^{EF}
0.75
1
Share of Workers with Hourly Pay, θ



$\overline{w}(\theta)$: type θ's hourly **reservation wage**, $\overline{w}(\theta) \equiv S^{-1}(\theta)$
MV(θ): **marginal value** of type θ's output, $MV(\theta) \equiv E[Y_i|\theta_i = \theta]$
AV(θ): **average value** among lower types, $AV(\theta) \equiv E[Y_i|\theta_i \leq \theta]$
$\overline{w}(\theta)$
MV(θ)
AV(θ)
Efficient Allocation
Welfare Loss
Competitive Equilibrium
Wage/Revenue ($)
0
0.25
θ^{EQ}
0.50
θ^{EF}
0.75
1
Share of Workers with Hourly Pay, θ
Monitoring Costs

Incorporating Moral Hazard

- Where does moral hazard fit in? Consider potential outputs

  Y_{1i}: Potential output value under hourly wage

  Y_{0i}: Potential output value under piece rate

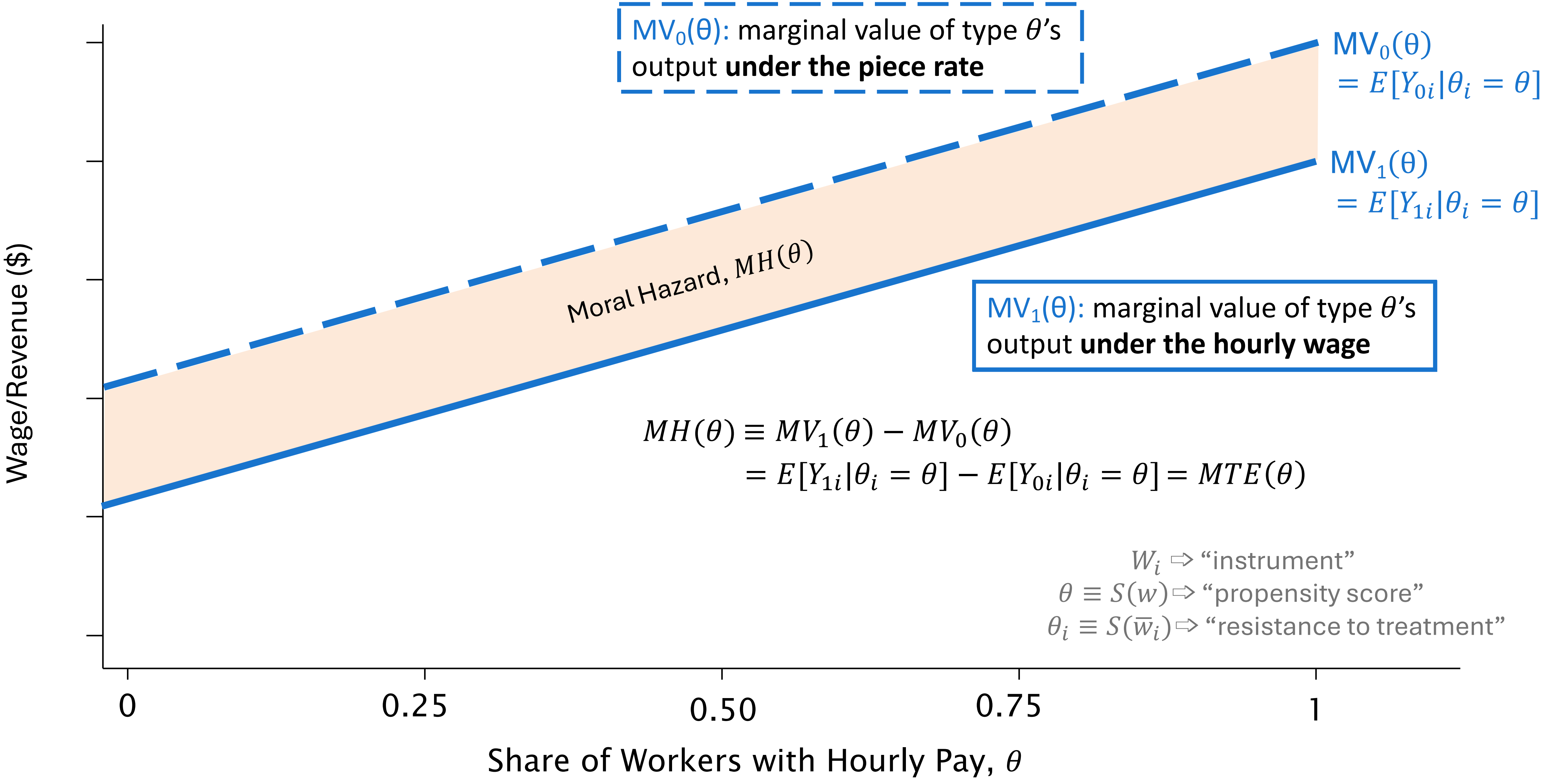
- Firms care about Y_{1i}, but they don't care about Y_{0i}
  - Piece-rate workers sell their output at a constant price per unit, so their productivity has no effect on firm profits

- $MV(\theta)$ & $AV(\theta)$ are defined *conditional* on accepting the hourly wage

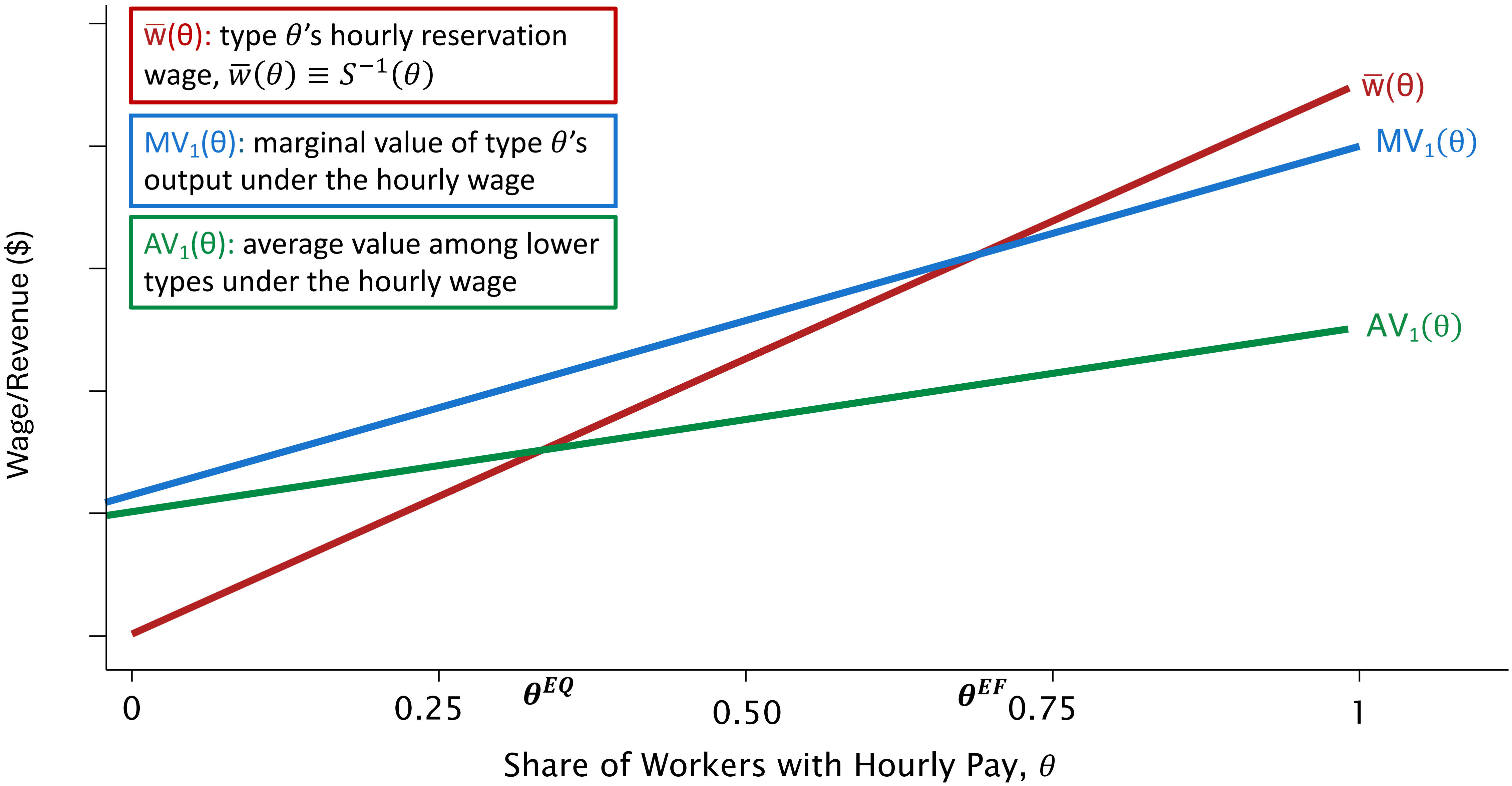
$$MV(\theta) \equiv E[Y_i|\theta_i = \theta] = E[Y_{1i}|\theta_i = \theta]$$
$$AV(\theta) \equiv E[Y_i|\theta_i \leq \theta] = E[Y_{1i}|\theta_i \leq \theta]$$

  $\Rightarrow$ Equilibrium is inclusive of workers' moral hazard response to the hourly wage

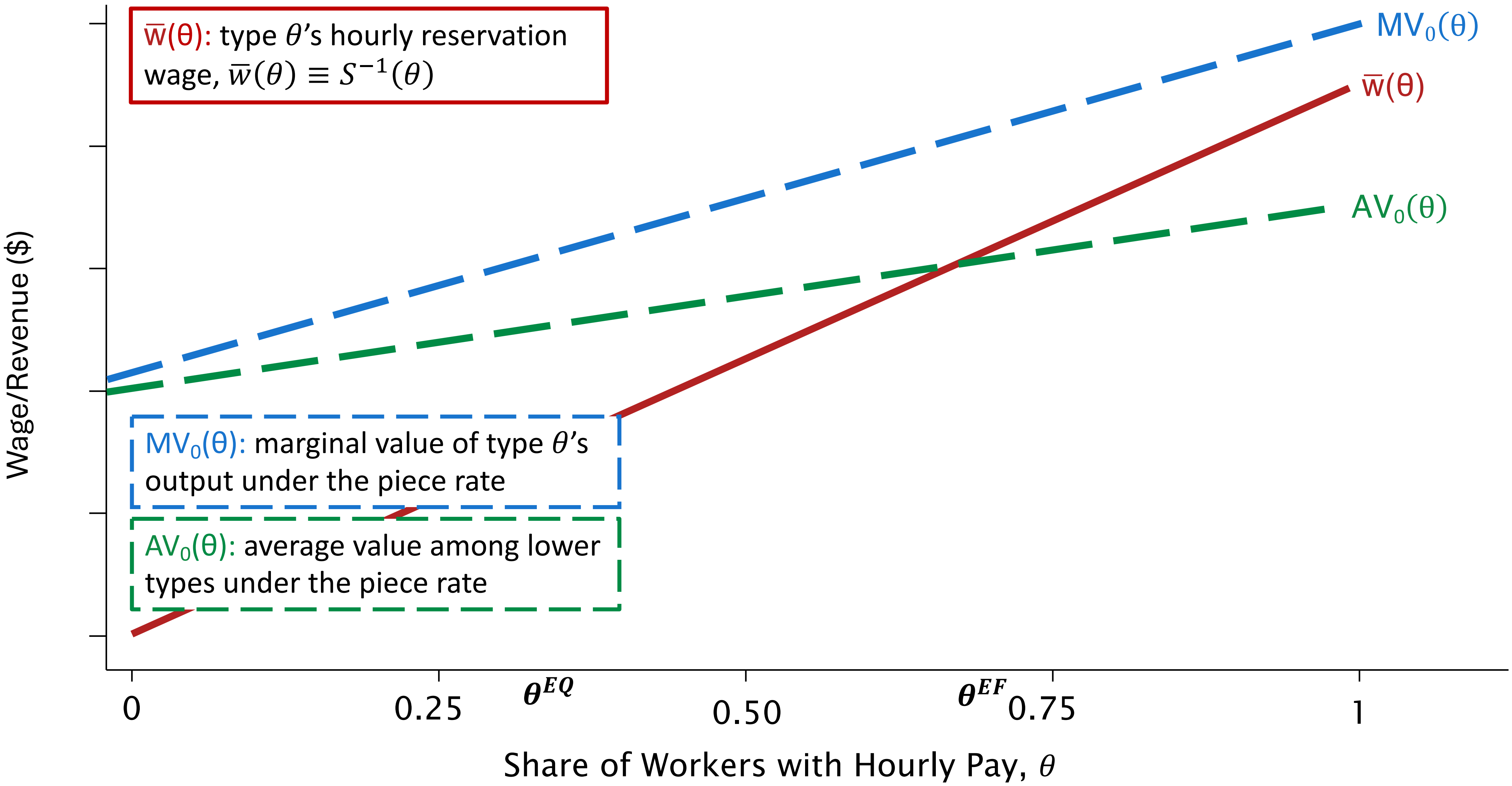
- Still want to separately identify moral hazard
  - Firms might mitigate MH with "partial insurance" (e.g., sales commission, tips)
  - Account for fiscal effects of decreased earnings under hourly wage subsidies



$MV_0(\theta)$: marginal value of type θ's output **under the piece rate**
$MV_0(\theta)$
$= E[Y_{0i}|\theta_i = \theta]$
$MV_1(\theta)$
$= E[Y_{1i}|\theta_i = \theta]$
Moral Hazard, $MH(\theta)$
$MV_1(\theta)$: marginal value of type θ's output **under the hourly wage**
Wage/Revenue ($)
$MH(\theta) \equiv MV_1(\theta) - MV_0(\theta)$
$= E[Y_{1i}|\theta_i = \theta] - E[Y_{0i}|\theta_i = \theta] = MTE(\theta)$
W_i ⇨ "instrument"
$\theta \equiv S(w)$ ⇨ "propensity score"
$\theta_i \equiv S(\overline{w}_i)$ ⇨ "resistance to treatment"
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay, θ



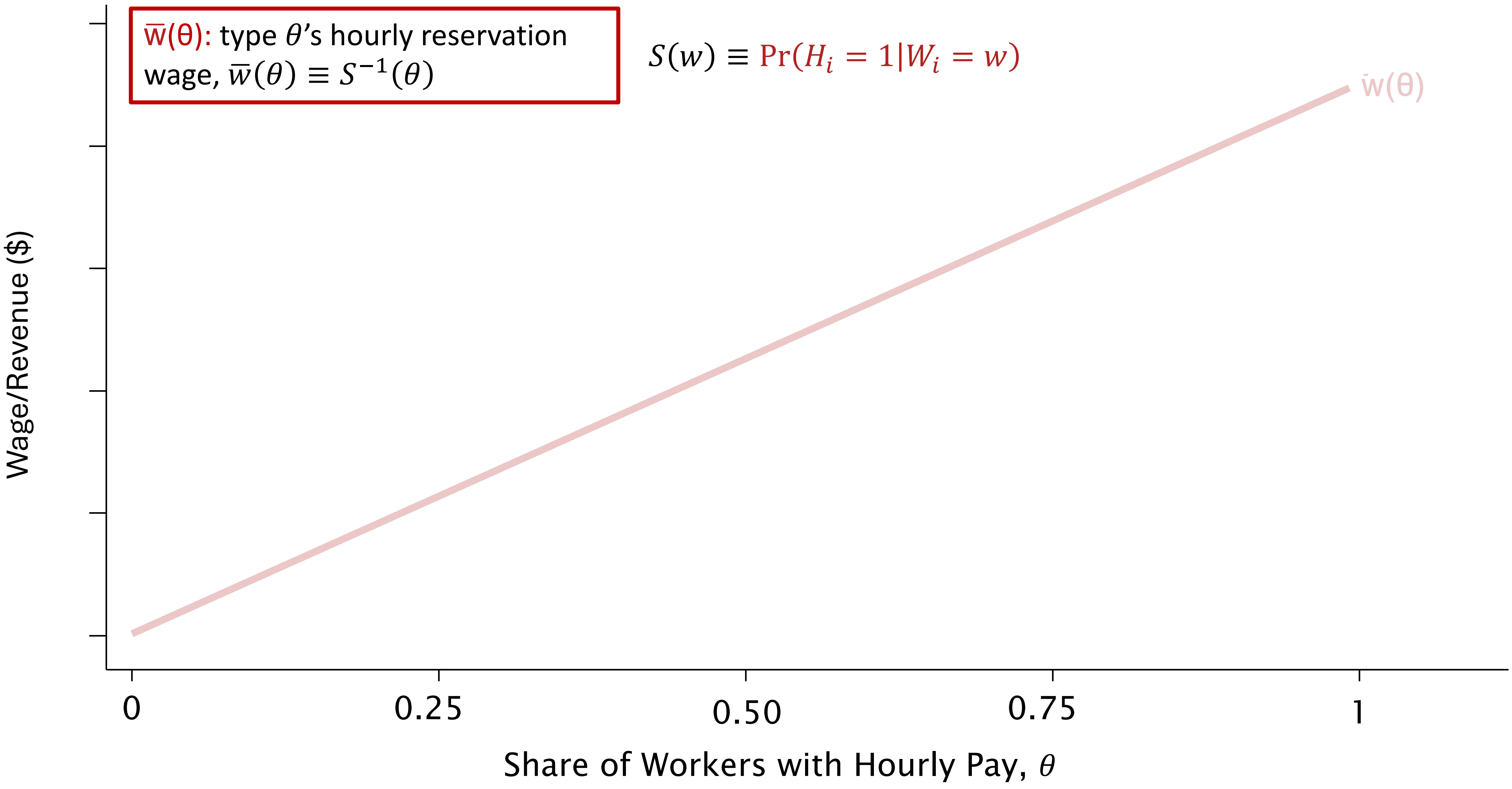
Monitoring Costs

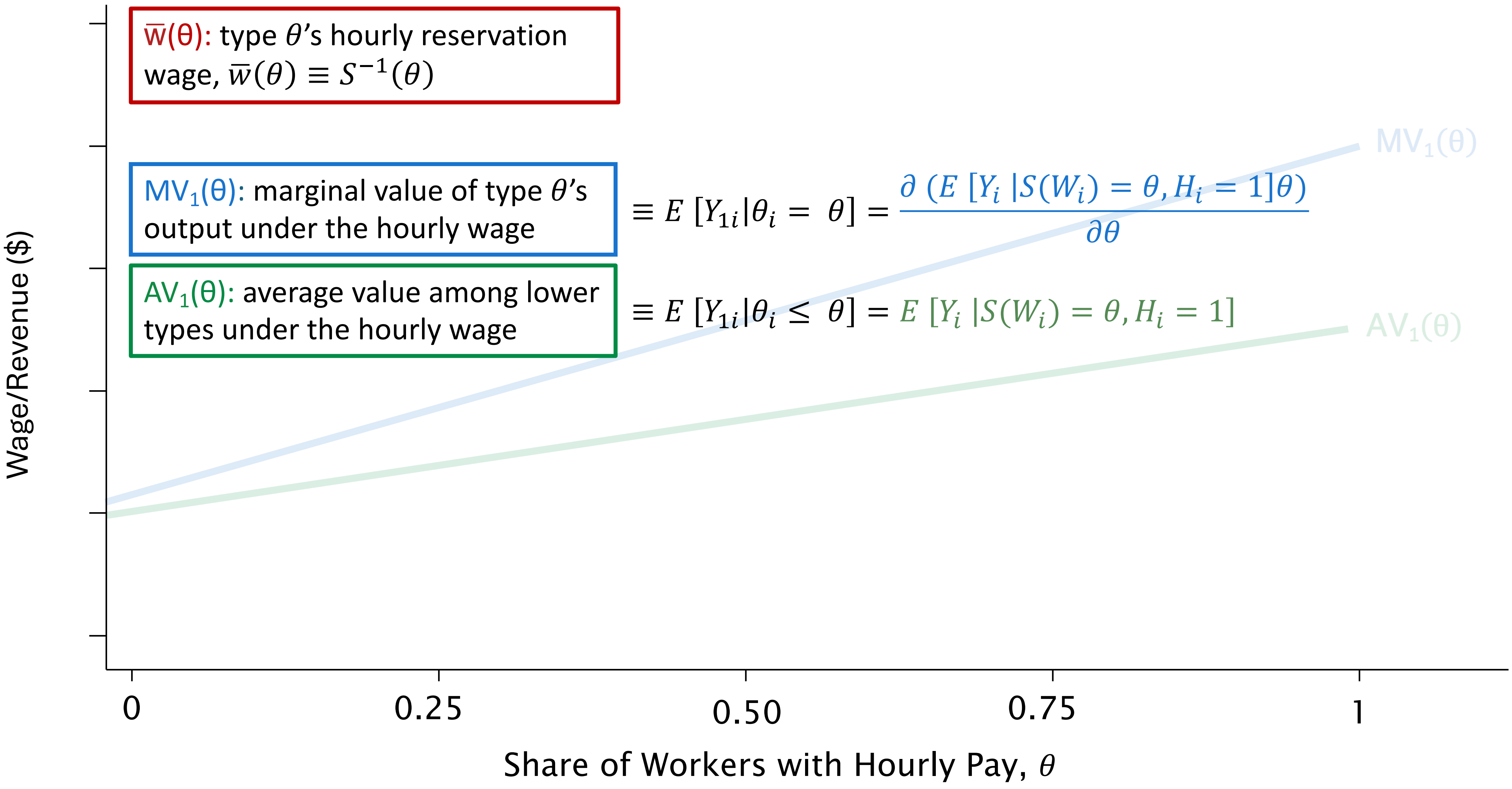


Monitoring Costs

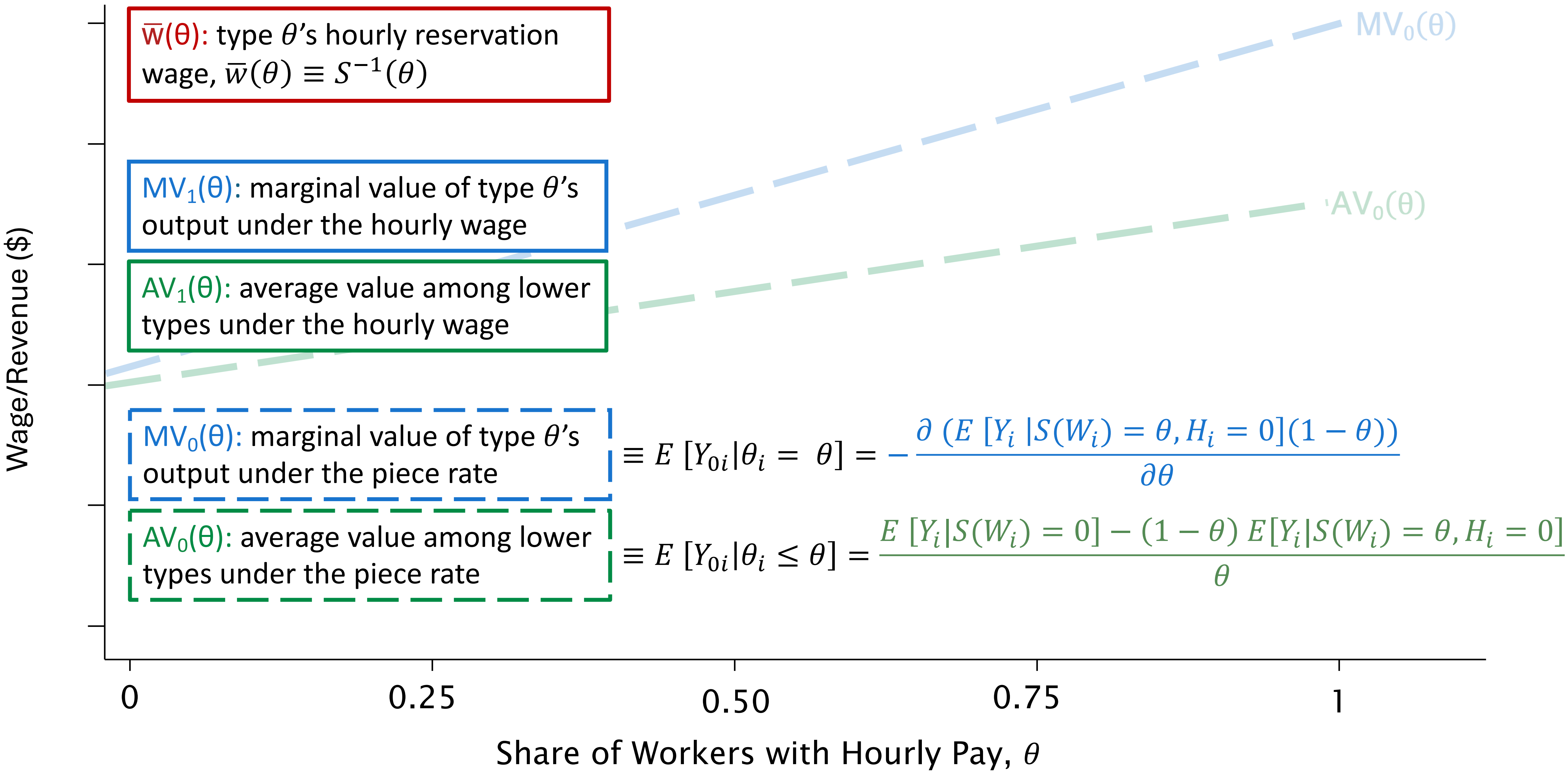
Outline

1. Experimental Design
2. Main Results
3. Model of Asymmetric Information
4. **Estimates of Marginal Value and Welfare Loss**
5. Optimal Wage Subsidies


$\overline{w}(\theta)$: type θ's hourly reservation wage, $\overline{w}(\theta) \equiv S^{-1}(\theta)$
$S(w) \equiv \Pr(H_i = 1 | W_i = w)$
$\overline{w}(\theta)$
Wage/Revenue ($)
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay, θ



$\overline{w}(\theta)$: type θ's hourly reservation wage, $\overline{w}(\theta) \equiv S^{-1}(\theta)$
$MV_1(\theta)$: marginal value of type θ's output under the hourly wage
$\equiv E\,[Y_{1i}|\theta_i = \theta] = \dfrac{\partial\,(E\,[Y_i\,|S(W_i) = \theta, H_i = 1]\theta)}{\partial\theta}$
$AV_1(\theta)$: average value among lower types under the hourly wage
$\equiv E\,[Y_{1i}|\theta_i \leq \theta] = E\,[Y_i\,|S(W_i) = \theta, H_i = 1]$
$MV_1(\theta)$
$AV_1(\theta)$
Wage/Revenue ($)
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay, θ


$\overline{w}(\theta)$: type θ's hourly reservation wage, $\overline{w}(\theta) \equiv S^{-1}(\theta)$
$MV_1(\theta)$: marginal value of type θ's output under the hourly wage
$AV_1(\theta)$: average value among lower types under the hourly wage
$MV_0(\theta)$: marginal value of type θ's output under the piece rate
$\equiv E\left[Y_{0i} | \theta_i = \theta\right] = -\frac{\partial \left(E\left[Y_i | S(W_i) = \theta, H_i = 0\right](1-\theta)\right)}{\partial \theta}$
$AV_0(\theta)$: average value among lower types under the piece rate
$\equiv E\left[Y_{0i} | \theta_i \leq \theta\right] = \frac{E\left[Y_i | S(W_i) = 0\right] - (1-\theta)\, E\left[Y_i | S(W_i) = \theta, H_i = 0\right]}{\theta}$
$MV_0(\theta)$
$AV_0(\theta)$
Wage/Revenue ($)
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay, θ

$\overline{\text{w}}(\theta)$: type θ's hourly reservation wage, $\overline{w}(\theta) \equiv S^{-1}(\theta)$

$$S(w) \equiv \Pr(H_i = 1 | W_i = w)$$

$MV_1(\theta)$: marginal value of type θ's output under the hourly wage

$$\equiv E\,[Y_{1i} | \theta_i = \theta] = \frac{\partial\,(E\,[Y_i\,| S(W_i) = \theta, H_i = 1]\theta)}{\partial\theta}$$

$AV_1(\theta)$: average value among lower types under the hourly wage

$$\equiv E\,[Y_{1i} | \theta_i \leq \theta] = E\,[Y_i\,| S(W_i) = \theta, H_i = 1]$$

$MV_0(\theta)$: marginal value of type θ's output under the piece rate

$$\equiv E\,[Y_{0i} | \theta_i = \theta] = -\frac{\partial\,(E\,[Y_i\,| S(W_i) = \theta, H_i = 0](1-\theta))}{\partial\theta}$$

$AV_0(\theta)$: average value among lower types under the piece rate

$$\equiv E\,[Y_{0i} | \theta_i \leq \theta] = \frac{E\,[Y_i | S(W_i) = 0] - (1-\theta)\,E[Y_i | S(W_i) = \theta, H_i = 0]}{\theta}$$

Estimation

$$S(w) \equiv \boxed{\Pr(H_i = 1 | W_i = w)}$$

1. Estimate $\Pr(H_i = 1 | W_i = w)$
   (logit regression)

2. Separately estimate

   $E[Y_i \,| S(W_i) = \theta, H_i = 1]$
   &
   $E[Y_i \,| S(W_i) = \theta, H_i = 0]$

   (local polynomial regression)

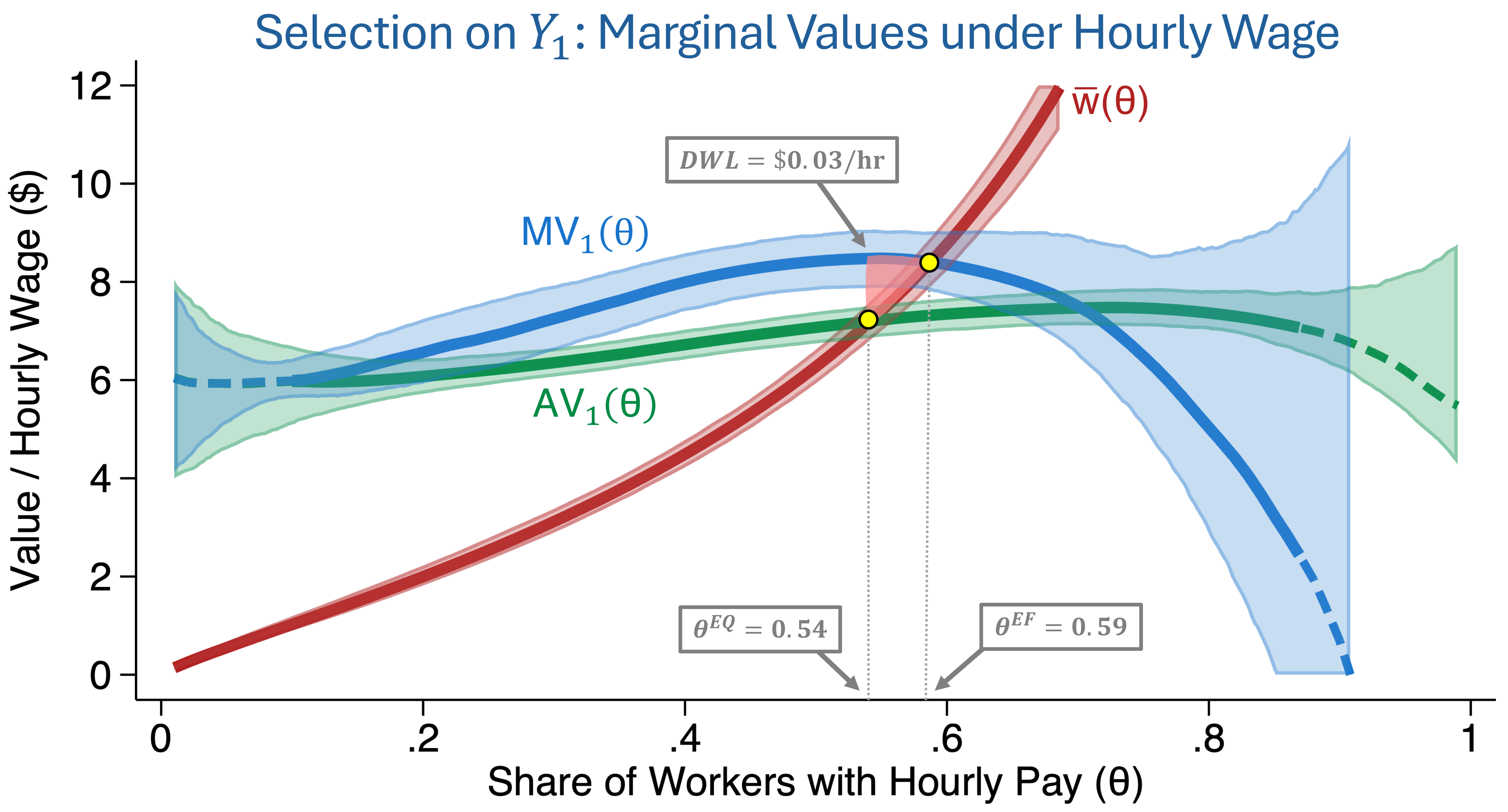
3. Differentiate with respect to θ

$$\mathrm{MV}_1(\theta) \equiv E\,[Y_{1i} | \theta_i = \theta] = \frac{\partial\,(E\,[Y_i \,| S(W_i) = \theta, H_i = 1]\,\theta)}{\partial\theta}$$

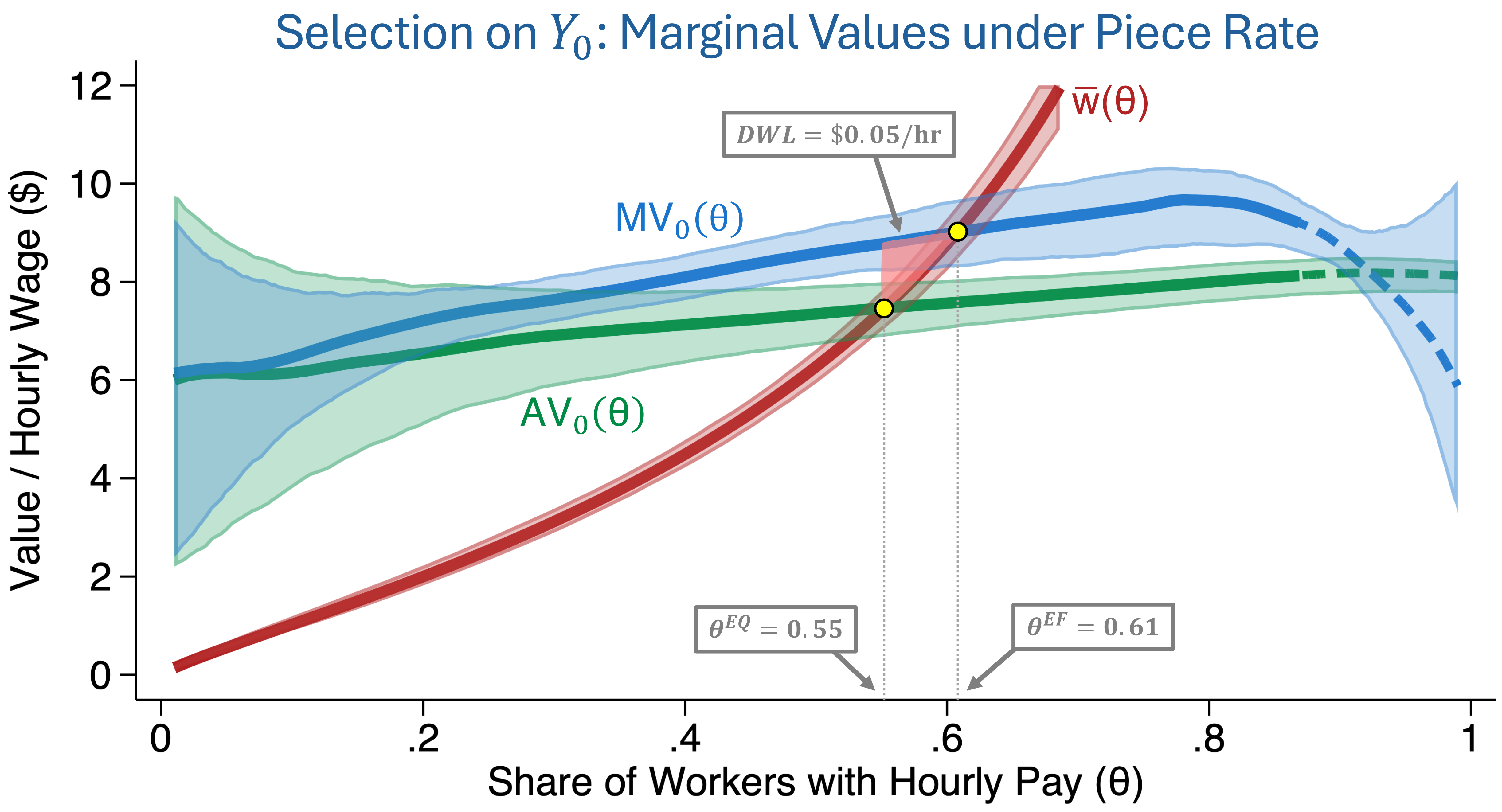
$$\mathrm{AV}_1(\theta) \equiv E\,[Y_{1i} | \theta_i \leq \theta] = \boxed{E\,[Y_i \,| S(W_i) = \theta, H_i = 1]}$$

$$\mathrm{MV}_0(\theta) \equiv E\,[Y_{0i} | \theta_i = \theta] = -\frac{\partial\,(E\,[Y_i \,| S(W_i) = \theta, H_i = 0](1 - \theta))}{\partial\theta}$$

$$\mathrm{AV}_0(\theta) \equiv E\,[Y_{0i} | \theta_i \leq \theta] = \frac{E\,[Y_i | S(W_i) = 0] - (1 - \theta)\,\boxed{E[Y_i | S(W_i) = \theta, H_i = 0]}}{\theta}$$

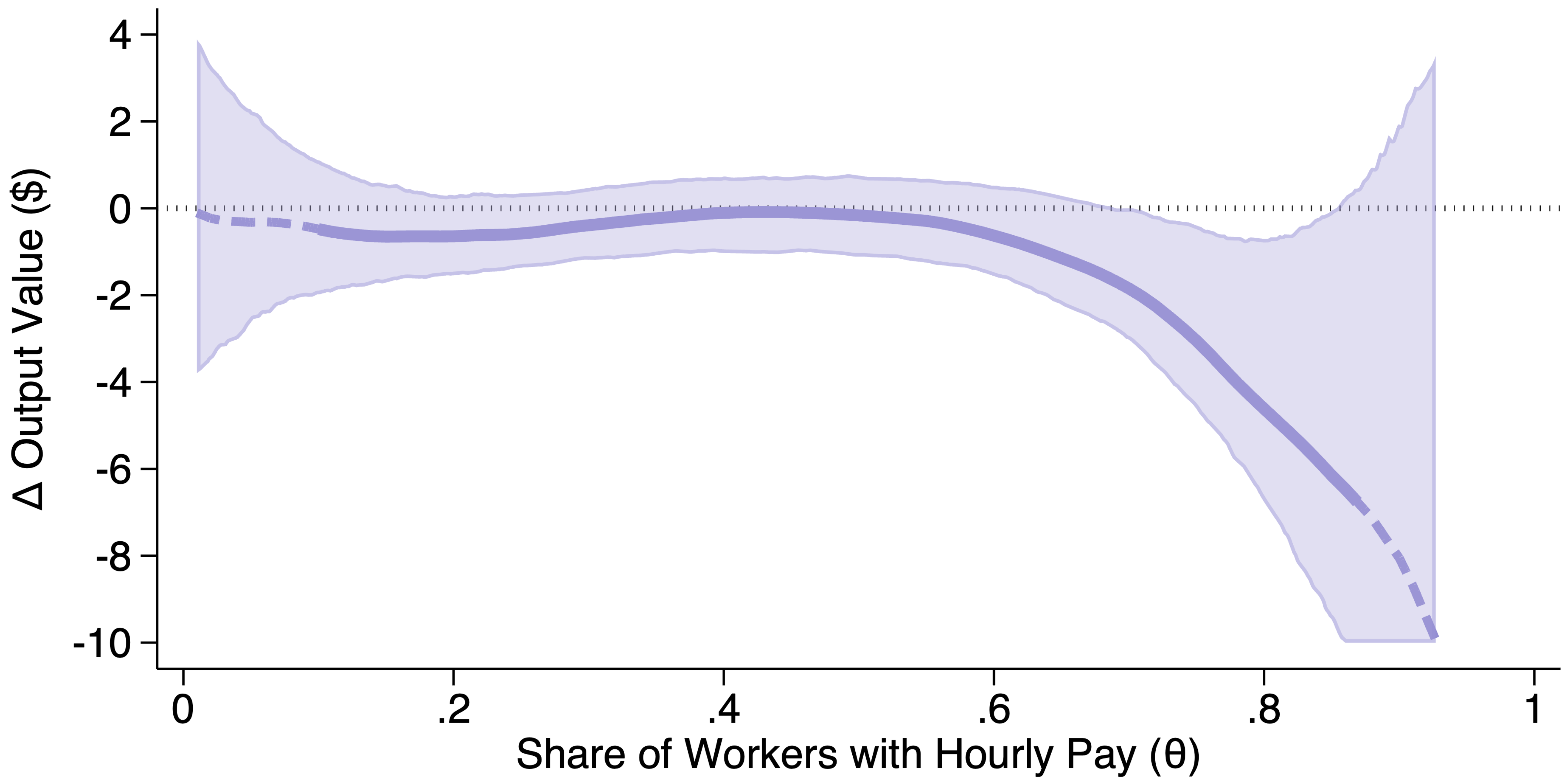


Selection on Y_1: Marginal Values under Hourly Wage
Value / Hourly Wage ($)
12
10
8
6
4
2
0
0
.2
.4
.6
.8
1
Share of Workers with Hourly Pay (θ)
$\bar{w}(\theta)$
$DWL = \$0.03/\mathrm{hr}$
$MV_1(\theta)$
$AV_1(\theta)$
$\theta^{EQ} = 0.54$
$\theta^{EF} = 0.59$



Selection on Y_0: Marginal Values under Piece Rate
Value / Hourly Wage ($)
Share of Workers with Hourly Pay (θ)
0
2
4
6
8
10
12
0
.2
.4
.6
.8
1
$\overline{w}(\theta)$
$DWL = \$0.05/\mathrm{hr}$
$MV_0(\theta)$
$AV_0(\theta)$
$\theta^{EQ} = 0.55$
$\theta^{EF} = 0.61$

Moral Hazard: Estimates of Marginal Treatment Effects



Outline

Policy Solutions?

Marginal Value of Public Funds (MVPF) for an hourly wage subsidy:

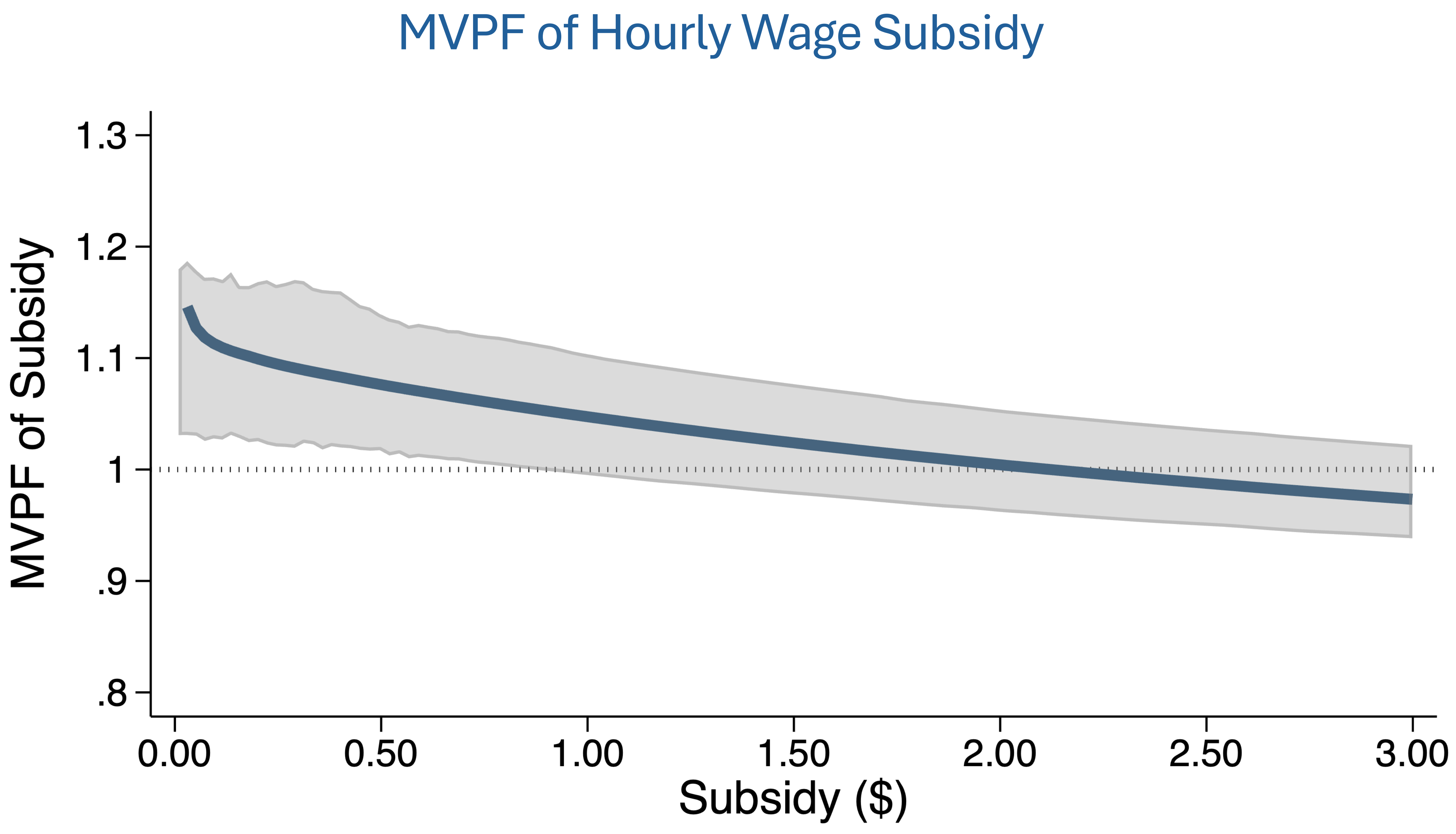
$$MVPF = \frac{Benefits}{Net\ Cost\ to\ Govt}$$

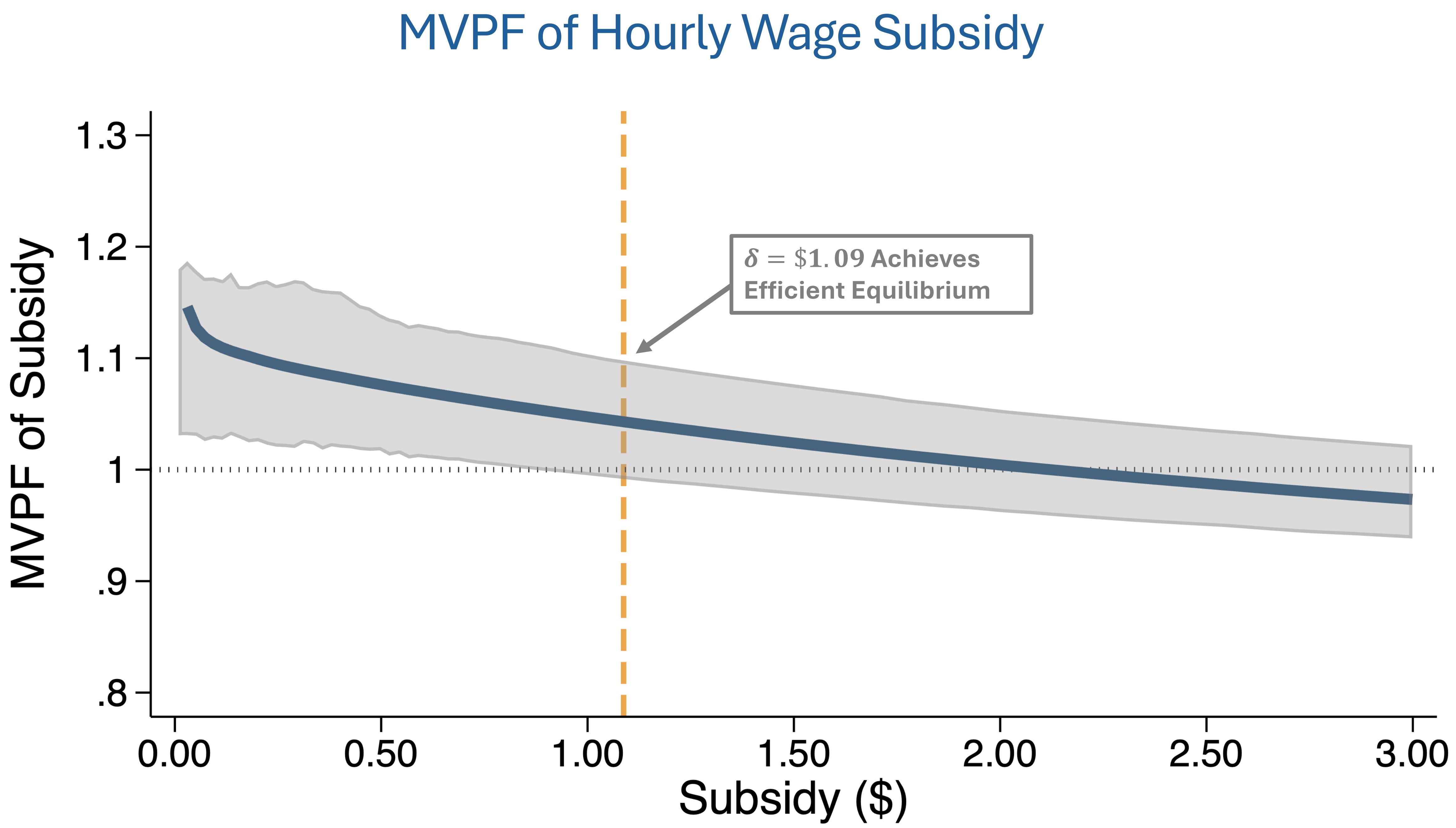
$Benefits$: The aggregate amount workers would be willing to pay for an hourly contract at the subsidized wage.

- \+ Net transfer from subsidy
- \+ Smoothing benefit from mitigating risk

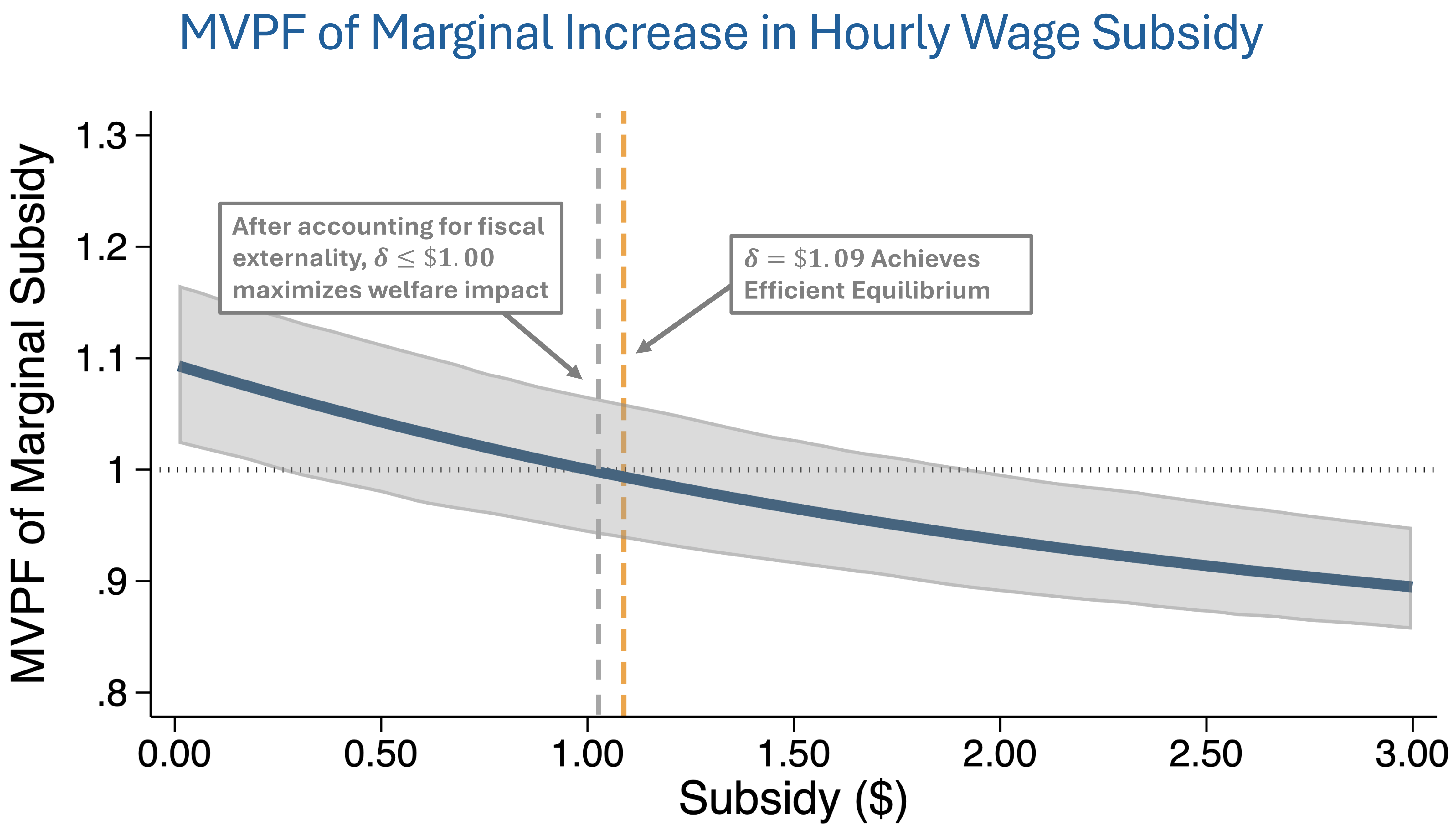
$Net\ Cost\ to\ Govt$: The aggregate amount spent, less program revenue or increased tax receipts

- − Net transfer from subsidy
- − Fiscal Externality: Reduced tax revenue from shirking workers


MVPF of Hourly Wage Subsidy
1.3
1.2
1.1
1
.9
.8
MVPF of Subsidy
0.00
0.50
1.00
1.50
2.00
2.50
3.00
Subsidy ($)

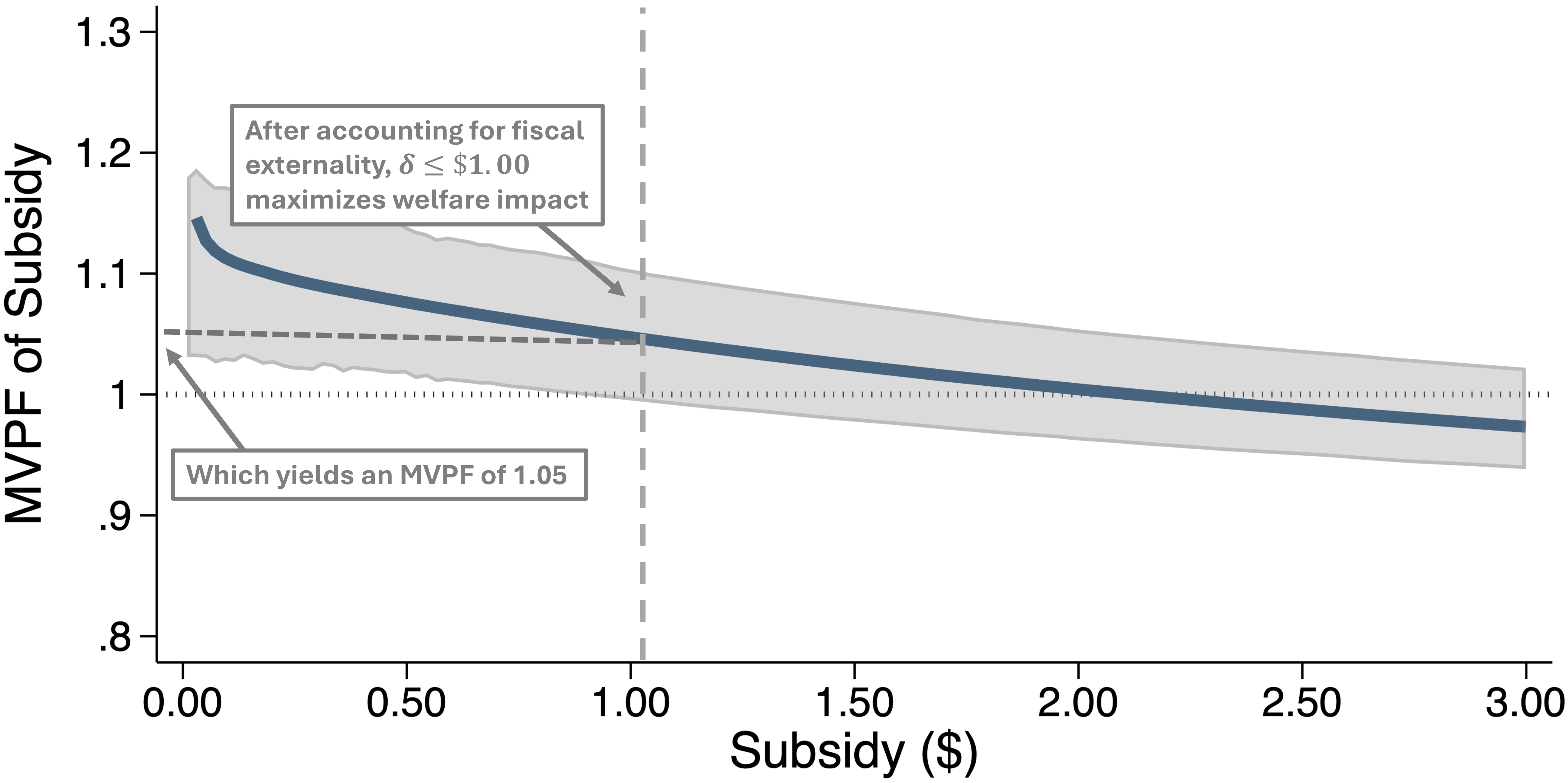


MVPF of Hourly Wage Subsidy
$\delta = \$1.09$ Achieves Efficient Equilibrium
MVPF of Subsidy
1.3
1.2
1.1
1
.9
.8
0.00
0.50
1.00
1.50
2.00
2.50
3.00
Subsidy ($)



MVPF of Marginal Increase in Hourly Wage Subsidy
After accounting for fiscal externality, $\delta \leq \$1.00$ maximizes welfare impact
$\delta = \$1.09$ Achieves Efficient Equilibrium
MVPF of Marginal Subsidy
1.3
1.2
1.1
1
.9
.8
0.00
0.50
1.00
1.50
2.00
2.50
3.00
Subsidy ($)

MVPF of Hourly Wage Subsidy



Conclusion

- Insurance model of labor contracts under asymmetric information
  - Shows how moral hazard and adverse selection affect market equilibrium
  - Parameters map into an MTE framework
- RCT estimates (marginal) selection and treatment effects of hourly pay
  - Separately identifies moral hazard and adverse selection
- Estimate welfare loss of $0.05/hour from inefficient contracts
  - Hourly wage subsidies of $1.00 or less can mitigate welfare loss
- Potential for distortions in whenever labor product is ex-ante unknown
- Future research and extensions
  - Employer Learning: screening on past performance
  - Soliciting preferences to estimate MH & AS in other settings or contract dimensions
    1. “What’s the minimum pay raise you would accept to forego your bonus?”
    2. Measure worker productivity by stated willingness-to-accept

Appendix

Identifying Selection Across Multiple Wage Offers

W_i: Randomized treatment-offer group

- $W_i = L$: Offered a choice between low hourly wage and piece rate
- $W_i = H$: Offered a choice between high hourly wage and piece rate
- π^W: Take-up of wage offer $W \in \{L, H\}$

Selection on Y_0:

$$E\left[Y_{0i} \middle| H_i^H = 1, H_i^L = 0\right] - E\left[Y_{0i} \middle| H_i^H = 0\right]$$
$$= \frac{1-\pi^L}{\pi^H - \pi^L} E[Y_i | H_i = 0, W_i = L] - E[Y_i | H_i = 0, W_i = H]$$

Selection on Y_1:

$$E\left[Y_{1i} \middle| H_i^L = 1\right] - E[Y_{1i} | H_i^H = 1, H_i^L = 0]$$
$$= \frac{\pi^H}{\pi^H - \pi^L} E[Y_i | H_i = 1, W_i = L] - E[Y_i | H_i = 1, W_i = H]$$

Back

Balance Test

	(1) Experimental Wage Offer	(2) Output Value
Number of Previous Tasks/1000	-0.0478	0.191^{***}
	(0.0343)	(0.0305)
Age	0.00141	-0.0683^{***}
	(0.00529)	(0.00453)
Female	0.0909	0.366^{***}
	(0.124)	(0.108)
Minority	-0.0528	-0.896^{***}
	(0.125)	(0.109)
Employed	-0.202	0.142
	(0.138)	(0.121)
Student	0.0685	-0.474^{***}
	(0.169)	(0.149)
F-statistic	1.019	36.492
p-value	0.426	0.000
N	3030	3030

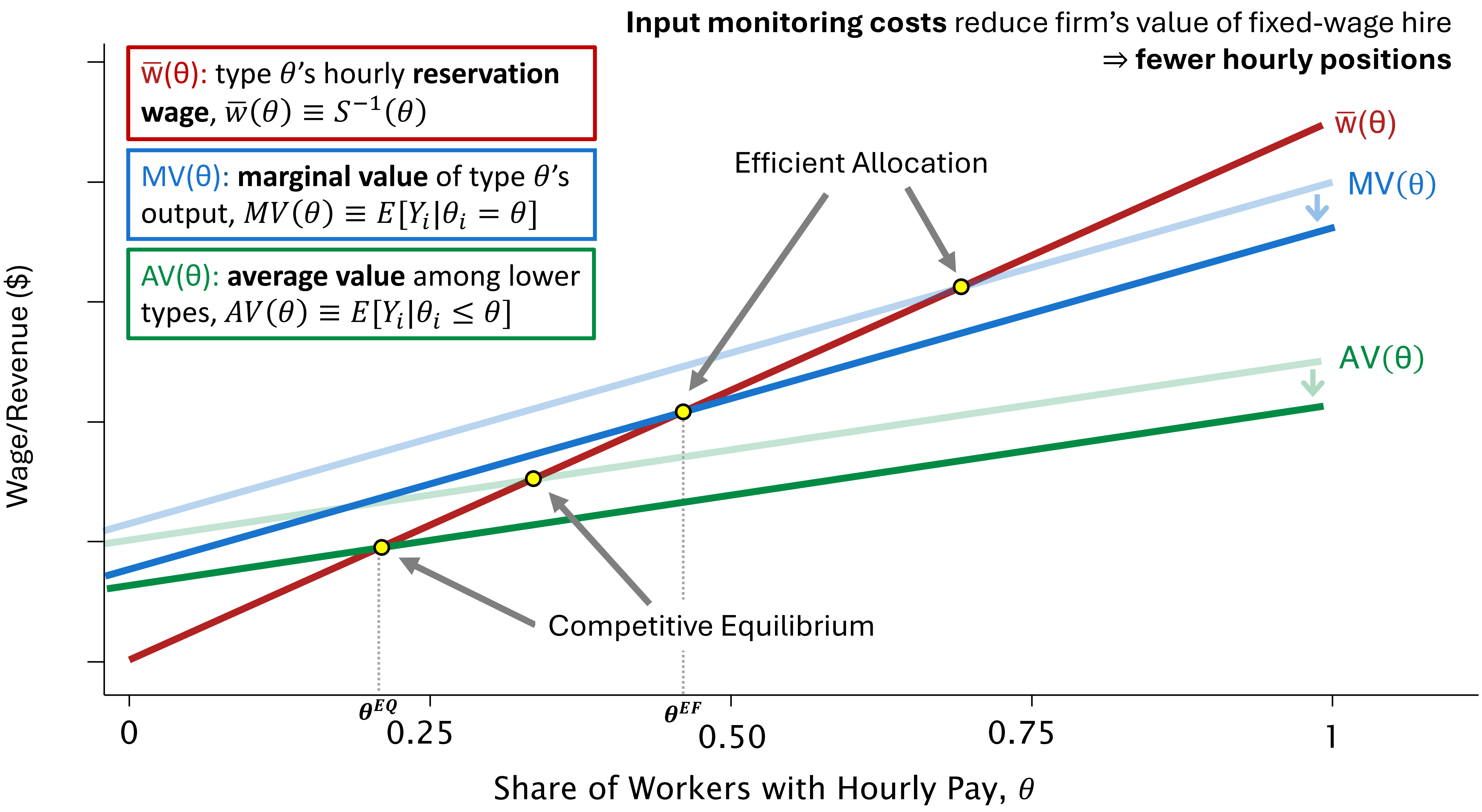
Back

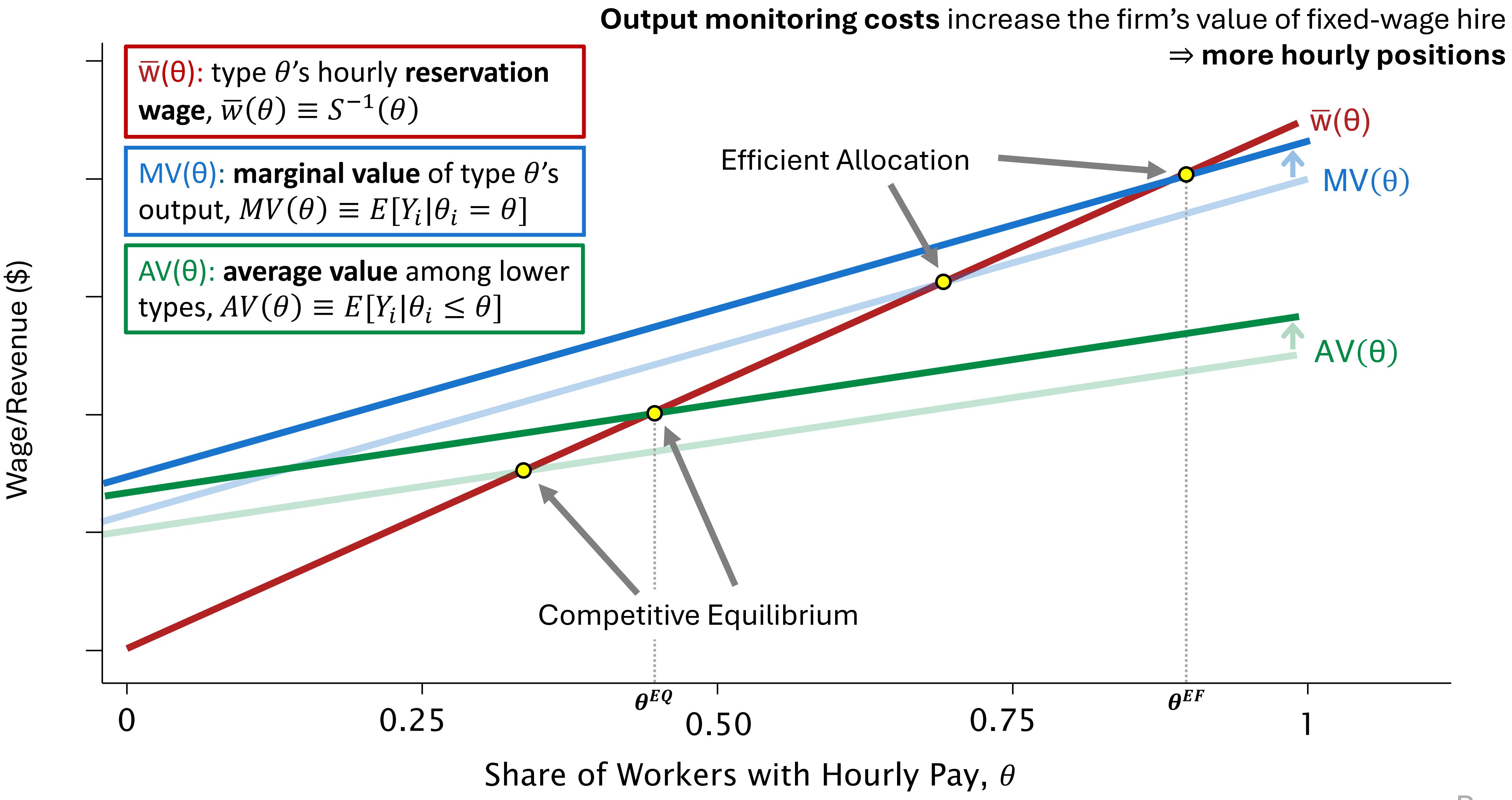
Hourly Wage Take-up

	(1) Accepted Offer	(2) Accepted Offer	(3) Accepted Offer	(4) Accepted Offer
Log Hourly Wage Offer	1.198***	1.202***	1.212***	1.245***
	(0.0554)	(0.0554)	(0.0560)	(0.0578)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
N	2728	2728	2728	2728

$$\Pr(H_i = 1) = \text{inv logit}\left(\alpha \ln W_i + \boldsymbol{\nu X_i}\right)$$

Main Results MTE


Input monitoring costs reduce firm's value of fixed-wage hire
⇒ **fewer hourly positions**
$\overline{w}(\theta)$: type θ's hourly **reservation wage**, $\overline{w}(\theta) \equiv S^{-1}(\theta)$
$MV(\theta)$: **marginal value** of type θ's output, $MV(\theta) \equiv E[Y_i|\theta_i = \theta]$
$AV(\theta)$: **average value** among lower types, $AV(\theta) \equiv E[Y_i|\theta_i \leq \theta]$
$\overline{w}(\theta)$
$MV(\theta)$
$AV(\theta)$
Efficient Allocation
Competitive Equilibrium
Wage/Revenue ($)
θ^{EQ}
θ^{EF}
0
0.25
0.50
0.75
1
Share of Workers with Hourly Pay, θ



Output monitoring costs increase the firm's value of fixed-wage hire
⇒ **more hourly positions**
$\overline{w}(\theta)$: type θ's hourly **reservation wage**, $\overline{w}(\theta) \equiv S^{-1}(\theta)$
MV(θ): **marginal value** of type θ's output, $MV(\theta) \equiv E[Y_i|\theta_i = \theta]$
AV(θ): **average value** among lower types, $AV(\theta) \equiv E[Y_i|\theta_i \leq \theta]$
Efficient Allocation
Competitive Equilibrium
$\overline{w}(\theta)$
MV(θ)
AV(θ)
Wage/Revenue ($)
0
0.25
θ^{EQ}
0.50
0.75
θ^{EF}
1
Share of Workers with Hourly Pay, θ

Back

Estimation Details

Estimation Follows Carneiro et al. (2012):

1. Estimate $S(w) \equiv \Pr(H_i = 1|W_i = w)$ using logit regression
2. Condition on X using double-residual regression (Robinson 1988)
   - X: Controls for number of previous tasks, task start time, and employment status
   - Assumes additive separability of X: $E[Y_{Ji}|X_i = x, \theta_i = \theta] = \xi_J \tilde{X}_i + MV_J(\theta)$
3. Separately estimate $E[Y_i|S(W_i) = \theta, H_i = 1]$ and $E[Y_i|S(W_i) = \theta, H_i = 0]$
   - Local polynomial regressions of $\tilde{Y}$ on $S(W_i)$ for hourly and piece-rate workers (bw=0.2)
4. Differentiate with respect to θ

Back